
TFE4130 WAVE PROPAGATION SOUND WAVES

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Chapter 1

Preface

This leaflet accompanies the lectures on sound waves in the course "TFE4130 Wave propagation". The text is written as if this part comes before the part on electromagnetic (EM) waves, so a number of fundamental concepts such as "What is a wave", "The $j\omega$ method", "Plane waves" etc are introduced here, and will be the same for EM waves.

In this leaflet, some central observations and statements are marked with boxes drawn with a thin line. A few numerical examples are given and they are marked by boxes drawn with a thick border line. There are also some sections that go beyond the scope of this course, and they are marked by boxes drawn with a dashed line, and the text starts with "Extra".

Chapter 2

Introduction to waves

2.1 What is a wave?

A wave is a disturbance/change of some physical quantity and this disturbance might propagate through space. Mathematically, this disturbance, which we can denote ξ , can be expressed as a function of space and time, $\xi(\mathbf{x}, t)$, where the position in space is represented by the 3 coordinates (x, y, z) , of a point, $\mathbf{x} = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z$, where $\mathbf{e}_x$ etc are the unit vectors in the cartesian coordinate system. One more mathematical aspect: the disturbance might be a single value, a *scalar*, for instance the sound pressure $p(\mathbf{x}, t)$, or the disturbance might have components in different directions, such as the displacement of solid media, $\mathbf{u}(\mathbf{x}, t) = u_x\mathbf{e}_x(\mathbf{x}, t) + u_y\mathbf{e}_y(\mathbf{x}, t) + u_z\mathbf{e}_z(\mathbf{x}, t)$. In this latter case, the disturbance is a *vector*.

We can observe waves around us as so-called *mechanical waves*, where there is a physical medium in which the physical quantity "resides" (a gas, a fluid, a string, a plate, solid material etc) or *electromagnetic (EM) waves*, which exist and propagate in vacuum, that is, no physical medium is needed to "house" the EM waves. Examples of waves are:

- *Surface waves in water*: we have all seen how a disturbance, e.g., from a boat passing, travels with quite a low speed across a lake.
- *Sound waves in air*: a sound signal will travel with $c = 343$ m/s (around 20°C) in air. Since we hear the same sound signal at different distances (just with different amplitudes), all frequencies travel with the same speed – and this indicates *non-dispersive wave propagation*. It is not immediately obvious, but *surface waves in water are dispersive* (for large water depths): waves of different frequencies travel with different speeds.
- *Sound waves underwater*: interestingly, underwater, a sound signal will travel with a much higher speed, $c \approx 1500$ m/s, than the surface waves

do. These waves are non-dispersive.

- *Vibrations on a string*: a movement/disturbance on a tight string will travel along the string, as a wave. The speed is quite high so we can not easily observe the wave propagation with our own senses. Furthermore, this type of wave is a *transversal wave* – the movement of the string is perpendicular to the direction of the propagation. Another not so obvious fact: sound waves in air and water are *longitudinal wave* – the air particles move in the same direction as the direction of propagation. Vibrations on an ideal string are non-dispersive.*
- *Seismics*: an earthquake generates vibrations in the interior of the earth (as well as along the surface), and these vibrations travel throughout the earth with a finite speed. Both longitudinal and transversal waves occur in seismics.
- *Electromagnetic (EM) waves*: a signal (light, or a radio wave, or other EM waves) will travel in vacuum with $c = 3 \cdot 10^8$ m/s.

We could also take a mathematical approach:

- Vibrations on a string: analysis will lead to a *partial differential equation* (PDE) for the vertical excursion. This turns out to be the one-dimensional *wave equation*, the very famous equation which was derived by d'Alembert in 1747. The solution to this differential equation will be the mathematical form of the observations we can make in reality.
- Sound waves: analysis will lead to a one-, two- or three-dimensional PDE for the so-called sound pressure. The PDE turns out to be the same PDE as for vibrations on a string – the wave equation.
- Electromagnetic waves: again we will encounter the same PDE – the wave equation (that comes out of Maxwell's equations).

EM waves are the main topic of this course, but we start with mechanical waves (where there is a physical movement of molecules/particles/physical objects), such as those listed above, since they have some easier properties than EM waves:

1. Sound waves in air are longitudinal waves, which leads to that they can be described with so-called *scalar wavefields*, rather than *vector wavefields*, which EM waves are. The math is a bit easier for scalar

*An ideal string, doesn't have any stiffness internally in the material, it is the tension caused by the mounting at the end points that causes the stiffness (which is necessary for a wave to be generated!). In reality, strings have some internal stiffness, and this non-ideal behaviour causes an increasing dispersion the higher the frequency is. This gradually appearing dispersion, in turn, leads to that musical string instruments have perfectly harmonic relationships between its frequency components for lower frequencies, and a gradual inharmonicity for higher frequencies.

wavefields than for vector wavefields.

2. The wave speed/propagation speed is so low for (some) mechanical waves that we can observe effects of the wave propagation quite concretely. We can for example hear the sound waves die out inside a reverberant space like a cathedral.
3. The simplest type of source for sound waves is a so-called *monopole source* (which generates a sound wave which is equally strong in all directions), but the simplest type of source for EM waves is a so-called *dipole source* (which has a $\cos \theta$ -radiation characteristic)^a. Again, the math of a monopole is a bit easier than for a dipole.

^aLongitudinal waves, and transversal waves that are scalar wavefields, can have monopole sources but vector wavefields, can not have monopole sources.

When we make a mathematical model of the physical reality, we often assume small signal amplitudes, and then the equations can become simpler. We can often assume *linear conditions*, when field quantities have small amplitudes. The wave equation for sound waves in air, and vibrations on a string, has been derived under such small-signal assumptions.

Linear conditions are enormously practical because *superposition* applies. Thanks to superposition, and to Fourier analysis, we can study a time-harmonic signal with just one (angular) frequency, ω . The result can then be used to derive how any transient signal will behave in the system we analyze.

2.2 Brief recap of the $j\omega$ method

If we have a “real-world” signal, $p_{\text{real}}(t)$, which contains a single angular frequency ω , then it can be written as

$$p_{\text{real}}(t) = \hat{p} \cos(\omega t + \varphi) \quad (2.1)$$

We introduce a complex version of this signal, $p_{\text{complex}}(t)$, such that the real-world signal is the real part of the complex version:

$$p_{\text{real}}(t) = \Re[p_{\text{complex}}(t)] \quad (2.2)$$

This complex version is (where we drop the subscript “complex”, so we just consider the complex version to be the standard one)

$$p(t) = Ae^{j\omega t} \quad (2.3)$$

where A is a complex constant. Comparing the real form with the complex one, we find the value of A :

$$p_{\text{real}}(t) = \hat{p} \cos(\omega t + \varphi) = \Re[\hat{p}e^{j(\omega t + \varphi)}] = \Re[\hat{p}e^{j\varphi}e^{j\omega t}] \quad (2.4)$$

So, the complex constant A in Eq. (2.3) is

$$A = \hat{p}e^{j\varphi}$$

It turns out that going from the simpler expression in Eq. (2.1) to the more-complicated-looking one in Eq (2.3), the mathematical operations become much easier. This becomes quite clear as soon as we try to add two or more sinusoids.*

2.3 What is a sound wave ?

We will study *sound waves* here, that is, movements in media like gases, fluids or solid media, that propagate through the medium. In solid media they are usually called *vibrations*, such as in strings, membranes, plates, the ground, etc: many music instruments generate vibrations in strings, which are transferred to vibrations in plates, which in turn generate sound waves in the air. It is the sound waves in the air that reach our ears so that we can hear the sound.

The quantity that we are primarily interested in is called the *sound pressure* $p(\mathbf{x}, t)$, [Pa], which is the small variation in air pressure, P_{total} , around the static/atmospheric air pressure (P_{atm}). The static air pressure can be viewed as the DC component of the total air pressure, and the AC component is called the sound pressure. First we assume that the sound pressure varies one-dimensionally, so that it is a function of x only,

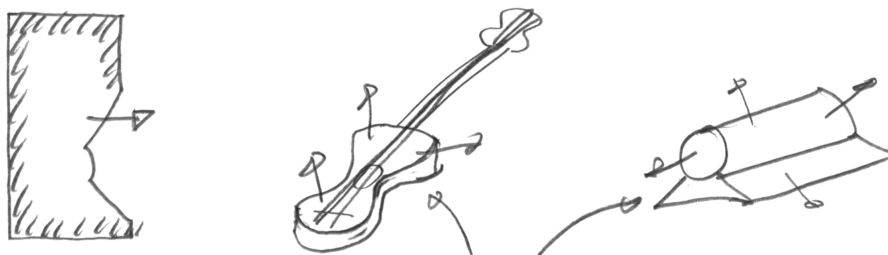
$$P_{\text{total}}(x, t) = P_{\text{atm}} + p(x, t)$$

If the sound pressure is a function of a single cartesian coordinate, then it is called a *plane wave*, which propagates in one direction only.[†]

Sound often originates from a vibrating surface. During the vibration, the surface moves outwards, "pushing" on the surrounding air particles, which leads to that the air pressure momentarily increases near the surface. The air pressure wants to return to P_{atm} , which leads to that the overpressure "pushes" on nearby air particles, which leads to a sound wave which propagates. If the vibrating surface is small, then the wave is propagating spherically (because the overpressure wants to expand in all directions), but for a large flat vibrating surface, a plane sound wave might be propagating. Figs. 2.1 and 2.2 show some examples of sound sources.

*The complex notation to describe waves was introduced by Lord Rayleigh in 1877.

[†]A plane wave could propagate in a single direction which does not coincide with one of the three cartesian coordinates, and then the sound pressure would be a function of two or three coordinates - but a rotation of the coordinate system would lead to a sound pressure which is a function of a single coordinate.



Complicated vibration patterns

Figure 2.1: Examples of sound sources with vibrating surfaces. To the left, a loudspeaker has a cone-shaped vibrating loudspeaker element mounted in a rigid box. In the middle, a violin has a wooden box the surface of which is vibrating with complicated vibration patterns which radiate sound. To the right, a general mechanical component, like an electrical motor, might also have more or less complicated vibration patterns on its surfaces, and these vibration patterns radiate sound.



Figure 2.2: Examples of sound sources with oscillating air being the source mechanism (arrows indicate the holes where the air is oscillating).

A short note on how we measure the sound pressure: a microphone is a transducer which transforms a sound pressure, at the location in space where we place the microphone, into an electrical signal, $e_{\text{mic}}(t)$. The microphone has a membrane which is pushed inwards when the air pressure at the membrane momentarily is higher than the static air pressure, P_{atm} , that is, when $p(x_{\text{mic}}, t)$ is positive, and the membrane is "sucked" outwards when the air pressure (momentarily) is lower than P_{atm} , that is when $p(x_{\text{mic}}, t)$ is negative.

Note 1: Both sides of the membrane are exposed to the static air pressure, so these two identical pressures don't move the membrane (except for the random variations inherent in air molecule movements), but the tiny sound pressure applies to one side only, which gives a net force, and this net force moves the membrane.

Note 2: Our ear functions in a similar way: the eardrum is constantly being exposed to the static air pressure on both the outside and the inside, and this keeps the eardrum balanced. Any change in total pressure on the outside (without a corresponding change on the inside of the eardrum), for instance as caused by an incoming sound wave, will give a net force which will move the eardrum.

This type of microphone is *omnidirectional*, which means that it will give the same output signal regardless of which direction a sound wave is coming from (as long as the microphone's membrane is much smaller than the wavelength of the incoming sound wave). This is possible because the sound pressure is a *scalar field*, as opposed to electromagnetic fields which generally are *vector fields*.

A last note about the sound pressure: it is indeed the sound pressure which is the "information-carrying" quantity when it comes to sound waves. The eardrum will get a vibration which is proportional to the sound pressure of the incoming sound wave.

Now we will do things a bit backwards: first we look at an animation of the solution, then we look at the mathematical form of the solution (to the wave equation) in Section 3.1, and last, we look at the derivation of the wave equation, in Section 3.2.

An animation of a plane wave will be played during the lecture. Fig. 2.3 shows a snapshot from that animation.



Figure 2.3: Snapshot of an animation of a plane wave, generated by a moving surface at the top.

Chapter 3

Plane sound waves

3.1 A plane sound wave

The simplest type of sound wave is the plane sound wave, illustrated in the animation in the previous section. This type of sound wave can be found in reality, e.g. when low-frequency noise is traveling inside a ventilation duct (which will act in the same way as a so-called waveguide for electromagnetic waves), or far away outdoors from any sound source. As argued earlier, we start by looking at a harmonic wave, that is, we assume that the sound wave has a sinusoidal time-dependence, with an angular frequency ω . Then the sound pressure for a plane sound wave which propagates in the positive x -direction can be written on the real-world form:

$$p_{\text{real}}(x, t) = \hat{p} \cos(\omega t - kx + \alpha) \quad (3.1)$$

where the so-called wavenumber $k = \omega / c$, where c is the speed of sound. α is a phase shift, which just depends on when we decide that $t = 0$ is. p obviously varies with x and t , and we can plot the sound pressure as function of x , for a given point in time, but also as function of t for a given point in space. This is illustrated in Fig. 3.1, where the similarity of ω and k is illustrated:

- ω tells how rapidly a wave is varying in the *time domain*.
- k tells how rapidly a wave is varying in the *space domain*.

However, we will see further below that the space domain has an important difference from the time domain: the wave number can have three components, k_x , k_y , and k_z , which tell how rapidly the wave is varying in the x -, y -, and z -direction, respectively. So, for the wave illustrated in Fig. 3.2, which

varies only as function of x , the wave number components would be

$$\begin{cases} k_x = k = \frac{\omega}{c} \\ k_y = 0 \\ k_z = 0 \end{cases}$$

To jump a little bit ahead, if a wave is varying as function of y and/or z as well, e.g., if a plane wave is propagating in some arbitrary direction, then the wavenumber components must be related as

$$k = \frac{\omega}{c} = \sqrt{k_x^2 + k_y^2 + k_z^2}$$

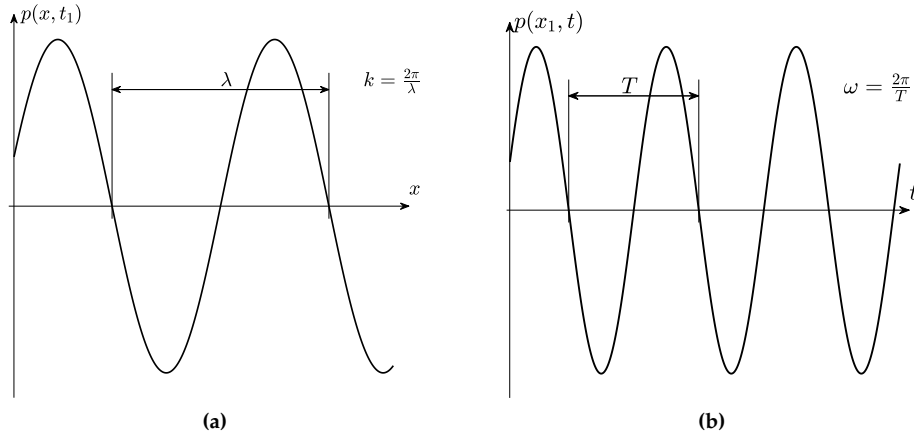


Figure 3.1: A sound wave, $p(x, t) = \hat{p} \cos(\omega t - kx + \alpha)$ plotted (a) as function of x , for a specific time instant t_1 , and (b) as function of t , for one specific location in space, x_1 .

We can inspect the expression in (3.1) a bit further: the function p has the same value if $\omega t - kx$ is constant. As a simplest first step, let's look at which x and t makes this constant = 0:

$$\omega t - kx = \text{const.} = 0 \Rightarrow x = \frac{\omega t}{k} = ct$$

This relationship implies that if we take snapshots of the sound pressure at times $t_1, t_2, t_3, \dots$, that is, we record (with many microphones) $p(x, t_i)$, and plot $p(x)$ for these consecutive times, then we will see that it looks like one part of the p -function will move and be in new positions, $x_i = ct_i$, that is, that part of the p -function will move towards larger x -values with a speed c .

Another form of animation of a plane wave will be played during the lecture.

Now we immediately take the step over to use the complex notation, and then the sound pressure for a plane wave that propagates in the positive x -direction is

$$p(x, t) = Ae^{j(\omega t - kx)} \quad (3.2)$$

As pointed out earlier, the two constants of the real form in Eq. (3.1), $\hat{p}$ and φ , are bunched together into the single complex amplitude A .

The form in Eq. (3.2) is a form which we will see again and again for representing a plane wave that propagates in the positive x -direction.^a

^aIn physics, it common to instead use a time-harmonic factor $e^{-i\omega t}$ (where i is used as the imaginary unit instead of j) and then a plane wave, propagating in the positive x -direction, has the form

$$p_{\text{physics}}(x, t) = Ae^{ikx} \cdot e^{-i\omega t} = Ae^{i(kx - \omega t)}$$

Both physicists and electrical engineers can be right! Convince yourself that

$$\Re[e^{j(\omega t - kx)}] = \Re[e^{i(kx - \omega t)}]$$

So, it is just a matter of convention/tradition, what one chooses. *But*, if one uses the expression $p(x) = e^{ikx}$, and attaches a time-harmonic factor $e^{i\omega t}$, then the result will be wrong (we get a wave that propagates in the opposite direction of what we intended).

One handy thing with the complex form is that the time-dependence will be identical in all expressions for waves, $e^{j\omega t}$, and can be factored out from all wave expressions. Often, a formulation like this is used in articles and books: *The time-harmonic factor $e^{j\omega t}$ is suppressed throughout the text. A plane wave, propagating in the positive x -direction has the form*

$$p(x) = Ae^{-jkx} \quad (3.3)$$

You will have to excuse the slight abuse of notation: p is used both for the function of x and t , and for the function of x , only, after we have removed the factor $e^{j\omega t}$.

We can see one example of the advantage of the complex notation. If we have several waves that propagate in the positive x -direction, with the same frequency but different amplitudes and phase shifts, then they can be added (thanks to superposition, which is thanks to linearity) into a single plane wave. The amplitude of this single plane wave is easy to find with the complex notation:

$$p_{\text{total}}(x, t) = A_1 e^{j(\omega t - kx)} + A_2 e^{j(\omega t - kx)} + A_3 e^{j(\omega t - kx)} \quad (3.4)$$

where we should remember that $A_1 - A_3$ are complex numbers. Then it is clear that we can write

$$p_{\text{total}}(x, t) = A_{\text{total}} e^{j(\omega t - kx)}, \quad \text{where } A_{\text{total}} = A_1 + A_2 + A_3 \quad (3.5)$$

that is, we just need to add the complex numbers. On the other hand, if

we had used the real-world formulations:

$$p_{\text{total,real}}(x, t) = \hat{p}_1 \cos(\omega t - kx + \varphi_1) + \hat{p}_2 \cos(\omega t - kx + \varphi_2) + \\ + \hat{p}_3 \cos(\omega t - kx + \varphi_3) = \hat{p}_{\text{total}} \cos(\omega t - kx + \varphi_{\text{total}})$$

it would have been much more cumbersome to find the values $\hat{p}_{\text{total}}$ and φ_{total} .

A plane wave which propagates in the *negative x-direction* has the form

$$p(x, t) = B e^{i(\omega t + kx)} \quad (3.6)$$

If we jump a little bit ahead (to Section 3.2), we can mention that the partial difference equation which we call the wave equation has, for the one-dimensional problem, two possible solutions, and they are exactly these two plane waves, in the positive and negative x -direction, respectively. The amplitudes A and B can be found by applying the *boundary conditions (BC)*, which specify, e.g., if there is a source somewhere, and/or a reflecting surface.

We can note that the amplitude isn't attenuated for the ideal, mathematical plane wave as it is propagating through air, (or through water). A real sound wave can behave pretty much like this because there is indeed little attenuation of a sound wave which propagates through air. In reality, however, there is *some* attenuation, which happens to be strongly frequency-dependent. Thus, low-frequency sound waves can be attenuated extremely little whereas high-frequency waves are attenuated much more. This is impractical for ventilation noise as low-frequency noise from the fan can propagate over long distances without being attenuated. Also outdoor noise can propagate very long distances thanks to this. Recently, the "world-record" was found when it comes to reverberation time, that is, how long a sound is audible before it dies out, inside a room. This world-record was found in an abandoned, enormous oil tank in Scotland. See:

<https://www.independent.co.uk/news/science/now-weve-heard-it-all-acoustic-scientists-shatter-the-world-record-for-longest-ever-echo-9062134.html>.

Also EM waves experience extremely little attenuation, and it is after all possible to receive signals from enormous distances in outer space.

The linearity, which implies that superposition is valid, has some very useful and not immediately obvious consequences. Two waves that cross each others' paths will continue as if the other didn't exist. See the top animation at: <https://www.acs.psu.edu/drussell/Demos/superposition/superposition.html> .

3.1.1 A plane sound wave in an arbitrary direction

A sound wave can, of course, propagate in any direction. Or, we can choose to orient our coordinate system in any direction, which has the same effect. We will look at three examples: a wave propagating parallel to the x -axis, as was studied above; a wave propagating parallel to the y -axis; and a wave propagating in a direction in the xy -plane, with a 30-degree angle to the x -axis, see Fig. 3.2a. We first plot the sound pressure as function of time for these three waves, see Fig. 3.2b. Apart from the (usually) irrelevant phase differences between the waves, their amplitudes are identical, and the three waves have the same period time, T , so the three waves must have the same angular frequency, ω (see Fig. 3.1b). Now, we can plot the sound pressure as function of x and y ,

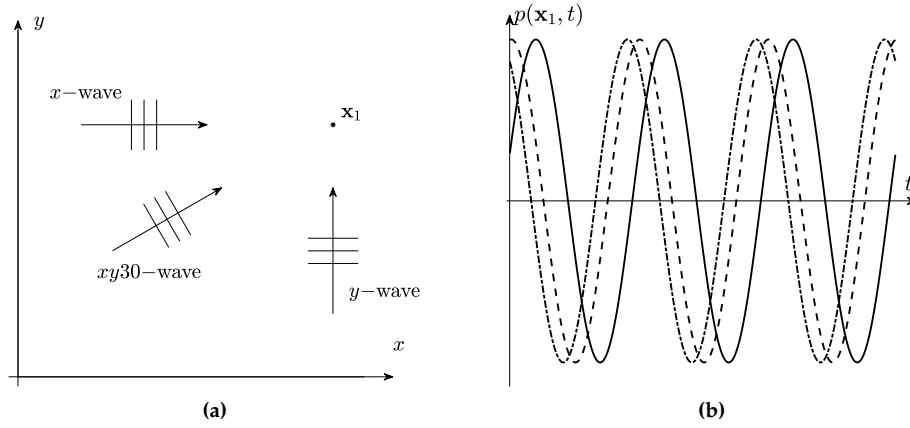


Figure 3.2: (a) Three plane waves, propagating in three different directions. (b) The sound pressure as function of time, for the three waves, at some arbitrary position in space, $\mathbf{x}_1$.

as shown in Figures 3.3, 3.4 and 3.6. For the x -wave, the sound pressure is identical for all value of y , because they all belong to the same wave-front, and the sound pressure varies identically over time. We can find the wavenumbers from the observed variation period in Fig. 3.3:

$$\begin{cases} k_x = \frac{\omega}{c} \\ k_y = 0 \end{cases} \quad \text{for the } x\text{-wave}$$

Obviously, it gets reversed for the y -wave as shown in Fig. 3.4:

$$\begin{cases} k_x = 0 \\ k_y = \frac{\omega}{c} \end{cases} \quad \text{for the } y\text{-wave}$$

It gets a bit more interesting for the $xy30$ -wave, because we can observe variation both as function of x and of y . Figs. 3.5 and 3.6 illustrate that we can observe a variation period along the x - and y -axis which is always longer than

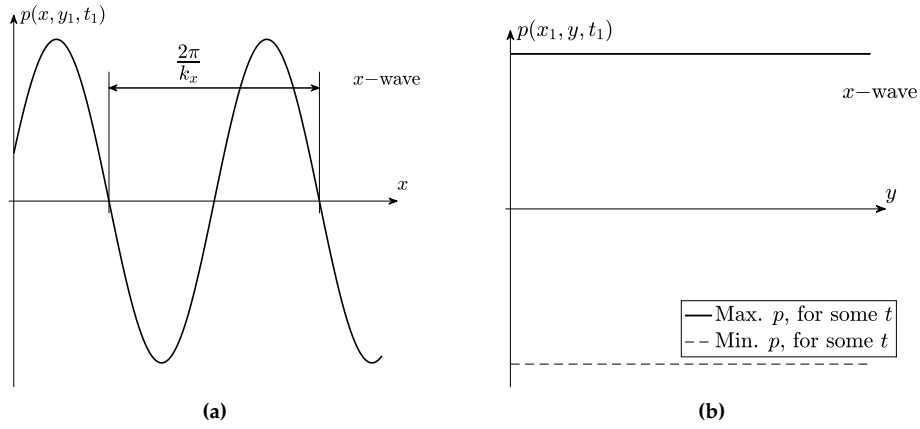


Figure 3.3: The sound pressure (a) as function of x , for some arbitrary value of y and t , and (b) as function of y , for some arbitrary value of x , and values of t which give the maximum and minimum values of p , for the x -wave in Fig. 3.2a.

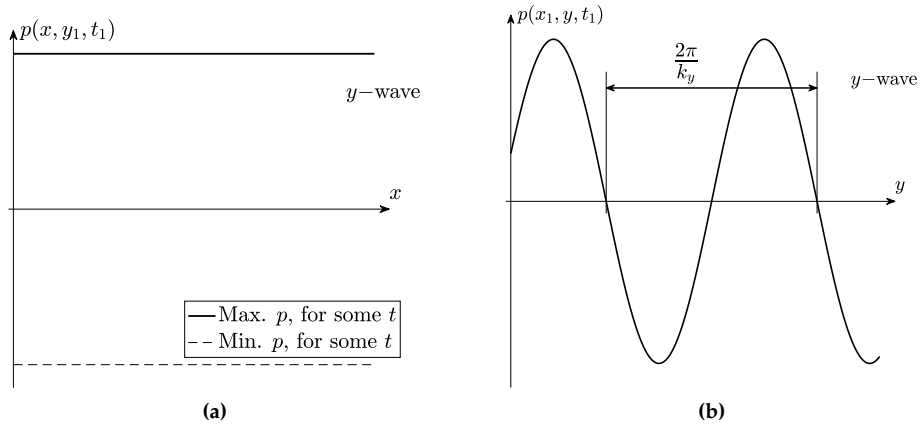


Figure 3.4: The sound pressure (a) as function of x , for some arbitrary value of y and t , and (b) as function of y , for some arbitrary value of x , and values of t which give the maximum and minimum values of p , for the y -wave in Fig. 3.2a.

the wavelength λ , and such observed variation periods along the x - and y -directions could be called *apparent wavelengths*, such as $\lambda_{\text{apparent},x}$ in Fig. 3.5. The wavenumbers can be found from these observed variation periods, or apparent wavelengths:

$$\begin{cases} k_x = \frac{2\pi}{\lambda_{\text{apparent},x}} = \frac{2\pi}{\lambda/\cos\theta} = \frac{\omega}{c} \cos\theta \\ k_y = \frac{2\pi}{\lambda_{\text{apparent},y}} = \frac{2\pi}{\lambda/\sin\theta} = \frac{\omega}{c} \sin\theta \end{cases}$$

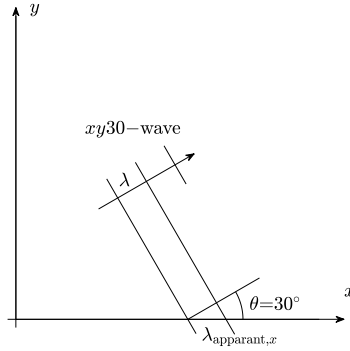


Figure 3.5: Wavefronts of the $xy30$ -wave in Fig. 3.2, illustrating that the observed variation along the x -axis will display an “apparent wavelength”, $\lambda_{\text{apparant},x}$ which is longer than the sound wave’s wavelength: $\lambda_{\text{apparant},x} = \lambda / \cos \theta$.

It should be realized that if we had created a rotated 2D-coordinate system (ζ, η) such that the $xy30$ -wave propagates along the ζ -axis, then

$$\begin{cases} k_{\zeta} = \frac{2\pi}{\lambda} = \frac{\omega}{c} = \frac{\omega}{c} \cos 0 \\ k_{\eta} = 0 = \frac{\omega}{c} \sin 0 \end{cases}, \text{ for the } xy30\text{-wave}$$

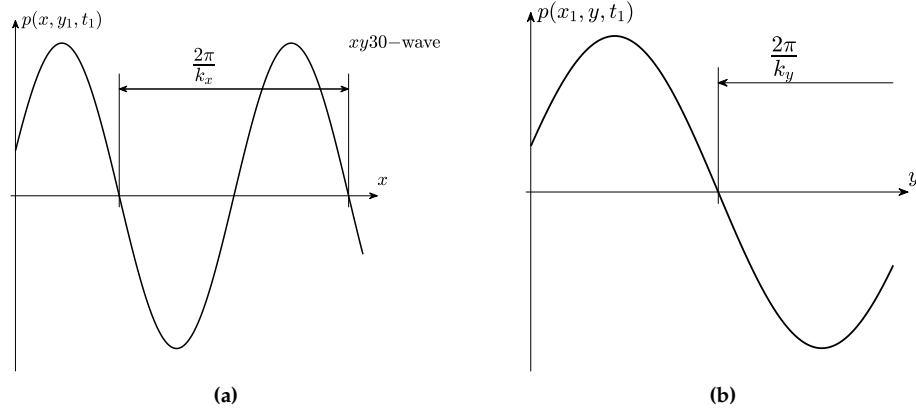


Figure 3.6: The sound pressure (a) as function of x , for some arbitrary value of y and t , and (b) as function of y , for some arbitrary value of x , and t , for the $xy30$ -wave in Fig. 3.2a.

A plane wave which propagates in the xy -plane, with an angle θ to the positive x -axis, can thus be written as

$$p(x, y, t) = Ae^{-jk_x x} e^{-jk_y y} e^{j\omega t} = Ae^{-j(k \cos \theta)x} e^{-j(k \sin \theta)y} e^{j\omega t}$$

$$= Ae^{-j\mathbf{k} \cdot \mathbf{x}} e^{j\omega t} \quad (3.7)$$

where the last, particularly compact, form expresses (x, y) and (k_x, k_y) as vectors:

$$\mathbf{x} = x\mathbf{e}_x + y\mathbf{e}_y = (x, y),$$

$$\mathbf{k} = k_x\mathbf{e}_x + k_y\mathbf{e}_y = (k_x, k_y).$$

where $\mathbf{e}_x$ and $\mathbf{e}_y$ are the unit vectors in the x - and y -directions, respectively.

The z -direction is straightforward to introduce as well, but one has to specify two angles, e.g., θ and φ , according to a spherical coordinate system:

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$$

To summarize, the wavenumbers in the three directions relate to the angular frequency as

$$\frac{\omega}{c} = k = \sqrt{k_x^2 + k_y^2 + k_z^2}$$

Extra In Fig. 3.5, we could imagine a rotated coordinate system, (ξ, η) , which results if we rotate the (x, y) coordinate axes by 30 degrees in the counterclockwise direction. Then the ξ -direction would coincide with the propagation direction of the $xy30$ -wave. A rotation of a cartesian coordinate system can be implemented with a rotation matrix:

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

The first line of this matrix equation tells us that

$$\xi = \cos \cdot \theta x + \sin \cdot \theta y$$

We can rewrite the plane wave expression in Eq. (3.7) as:

$$p(x, y, t) = Ae^{-jk(\cos \cdot \theta x + \sin \cdot \theta y)} e^{j\omega t}$$

and recognize the coordinate transformation from above, so a single-coordinate plane wave results:

$$p(\xi, t) = Ae^{-jk\xi} e^{j\omega t}$$

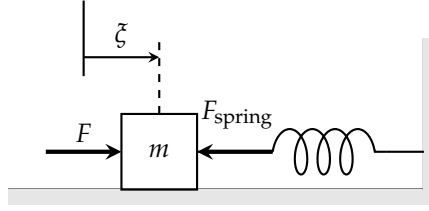


Figure 3.7: A mass-spring system

3.2 The 1D wave equation

Above we saw the solutions for the sound pressure in a plane sound wave, Eqs. (3.2) and (3.6). Now we will look at where that solution comes from. We will have to derive the wave equation in 1D = the partial differential equation (PDE), which describes how sound waves propagate in air and fluids, and also, as it happens, for vibrations on a string or on a membrane. You will later see the same wave equation for electromagnetic waves.

We start with an analogy. You might have seen the differential equation which describes the vibration of mass-spring system. A mass, m , is resting on the ground and can move horizontally without friction. It is attached to a wall via a spring with the spring constant s . A force, F , is applied to the mass, and then the mass is displaced by the amount ζ , see Fig. 3.7.

We can set up a force balance equation, using Newton's second law: the sum of external forces (= the net force), on a mass gives an acceleration of the mass:

$$m \frac{d^2 \zeta}{dt^2} = F_{\text{net}} \quad (3.8)$$

The net force is a sum of the external forces, taking the directions into account (the spring force is given by Hooke's law):

$$F_{\text{net}} = F - F_{\text{spring}} = F - s\zeta \quad (3.9)$$

Combining these two equations, we get a differential equation:

$$m \frac{d^2 \zeta}{dt^2} + s\zeta = F \quad (3.10)$$

3.2.1 The propagation of a sound wave

The same force equation as above can be applied to sound waves. The mass of the air particles represents the mass of the system above. Forces act between air particles similar to that of a spring (when we press an inflated bicycle tyre, we experience that compressed air acts as a spring). Therefore we can use the same force balance as above. We start by stating/claiming/assuming that there is a sound pressure that changes with time, and that it can also be different in different positions: $p = p(x, t)$. We can then see where the mass and spring laws take us.

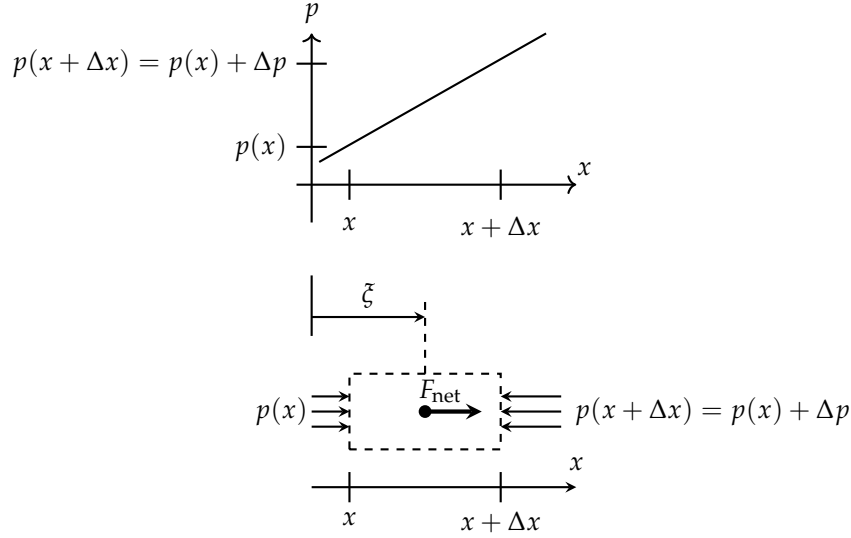


Figure 3.8: A small volume of air, with the length Δx in the x -direction, where two slightly different pressures act on the left side, $p(x)$, and the right side, $p(x + \Delta x)$, respectively. The pressure difference leads to a net force which acts on the volume of air.

- *The mass law for a small volume of air particles*

Consider a one-dimensional sound wave $p = p(x, t)$. We look at two different positions; x and $x + \Delta x$, see Fig. 3.8. The total air pressures in these two positions are slightly different, but by definition, the static air pressure is the same everywhere, so we need to inspect only the difference in *sound pressures*. If we look at the two positions at the exact same moment (i.e. at the same t) we get sound pressures $p(x)$ and $p(x + \Delta x)$. We can call the latter $p(x) + \Delta p$, that is, we denote the small difference in sound pressure Δp .

We then realize that this difference causes a net force which will try to move the air particles between these two positions, i.e., between x and $x + \Delta x$. That net force, F_{net} , is

$$F_{\text{net}} = -S\Delta p$$

The negative sign is caused by the fact that a positive Δp tries to move the air particles to the left, i.e., in the negative x -direction. This net force will try to move that little volume of air by an amount ξ . So, Newton's 2nd law will tell us what will happen (the little imagined box of air has a cross-section area of S , so the volume of the box is $S\Delta x$, and $m = \rho_0 V = \rho_0 S\Delta x$, where ρ_0 is the density of air at rest, that is, when there is no sound, that is, no air pressure variation):

$$m \cdot a = m \frac{\partial^2 \xi}{\partial t^2} = F_{\text{net}}$$

$$\Rightarrow \rho_0 S \Delta x \frac{\partial^2 \xi}{\partial t^2} = -S \Delta p \Rightarrow \frac{\Delta p}{\Delta x} = -\rho_0 \frac{\partial^2 \xi}{\partial t^2} \quad (3.11)$$

For infinitesimally small volumes, this differential becomes:

$$\frac{\Delta p}{\Delta x} \rightarrow \frac{\partial p}{\partial x} = -\rho_0 \frac{\partial^2 \xi}{\partial t^2} = -\rho_0 \frac{\partial v_x}{\partial t} \quad (3.12)$$

where v_x is the *particle velocity* in the x -direction, $v_x = \frac{\partial \xi}{\partial t}$.

The particle velocity is thus given as

$$v_x = -\frac{1}{\rho_0} \int \frac{\partial p}{\partial x} dt \quad (3.13)$$

and, for the special (but very common) case of a time-harmonic signal, for which $\frac{d}{dt} = j\omega$,

$$v_x = -\frac{1}{j\omega\rho_0} \frac{\partial p}{\partial x} \quad (3.14)$$

Here we did some assumptions that might not be obvious:

- We inserted the density of air at rest, ρ_0 , but in reality, the density changes at the same time as the sound pressure changes: $\rho_{\text{total}}(x, t) = \rho_0 + \rho(x, t)$. If we assume small amplitudes, then $|\rho(x, t)| \ll \rho_0$ and this assumption is valid. A similar relationship can be observed between the sound pressure amplitude and the static air pressure, and we are for now interested in small amplitudes so that $|p(x, t)| \ll P_{\text{atm}}$. For most practical acoustical problems, the sound pressure amplitudes are indeed much, much smaller than the static air pressure, which reflects that this small-amplitude assumption is perfectly fine. Exceptions: when shock waves from explosions are studied; or powerful sound sources such as jet engines; or other very powerful sound sources.

In addition: when this small-amplitude assumption is not valid, and we have to introduce the variation of the density, it turns out that one gets products of $e^{j\omega t}$ -factors, which will generate twice the frequency, and higher harmonics. For very powerful loudspeakers, this could mean that an input signal of a single frequency will generate a radiated sound wave with harmonic overtones, which is called harmonic distortion, and this is unwanted for a sound reproduction like a loudspeaker.

- We ignored that there will be losses when a sound wave propagates through air. In reality, there are losses which increase with the square of the frequency. This implies that losses are very low at low frequencies.

- *The gas law gives us the spring-effect of the air*

We need some sort of "spring-law", i.e. a relation between force (sound pressure) and displacement. Note! We're interested in the dynamic relation; if we vary the displacement harmonically (like a sinusoidal) with a given excursion, what is then the harmonic variation in air pressure (remember: this variation is our so-called sound pressure)? We will start with a derivation which seems straightforward and simple, but which is incorrect, method 1, and then we will do the derivation more correctly, method 2. The first approach was believed to be correct until the development of the field of thermo-dynamics.

1. *Method 1 - simplified too much!*

The following seems reasonable: If we have an isolated air quantity with a given static air pressure P_{atm} , and we increase the volume with 1%, the pressure should decrease by 1%, right?

We can describe this by equations: if we increase an air volume from V to $V + \Delta V$, the pressure will decrease from P_{atm} to $P_{\text{atm}} - p$ (remember: we use p to describe a small change in the total air pressure). Our logical guessing is in other words:

$$\frac{p}{P_{\text{atm}}} = -\frac{\Delta V}{V} \Rightarrow p = -P_{\text{atm}} \frac{\Delta V}{V} \quad (\text{WRONG!})$$

If we consider one-dimensional compression we can replace $\Delta V/V$ with $\partial \xi / \partial x$, so that

$$p = -P_{\text{atm}} \frac{\partial \xi}{\partial x} \quad (\text{WRONG!}) \quad (3.15)$$

This all seems straightforward and logical. But, what's wrong? Well, we don't consider the fact that both the pressure and the temperature change when we compress an air volume. That's why we have to perform a more accurate analysis with the gas law, as follows.

2. *Method 2 - using the gas law*

The gas law gives the relation between pressure, temperature and volume:

$$P_{\text{total}} V = \text{const}_1 T$$

where const_1 is some constant. If we have a so-called *adiabatic process* (no heat transfer), it can be shown that the gas law gives

$$P_{\text{total}} V^\gamma = \text{const}_2, \quad \text{and } \gamma = 1,43 \text{ for air.} \quad (3.16)$$

where const_2 is another constant. Sound waves in air are normally adiabatic processes: p changes so quickly that no significant heat transfer happens, see Fig. 3.9.

So, now we would like to know the relation between a relative change of P_{total} and a relative change of V , i.e. between small changes in the

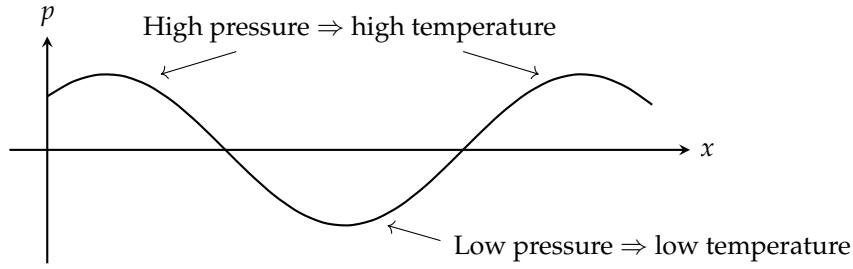


Figure 3.9: A snapshot of a sound wave. Note that “High” and “Low” mean very small changes from the values at rest. The heat conduction of air is so low that there is no time for the high temperature region to cause a heat transfer to the low temperature region - so an adiabatic process results.

total air pressure, $dP_{\text{total}} = p$, and the volume, dV . Eq. (3.16) implies that

$$\begin{aligned} P_{\text{total}} &= \text{const}_2 V^{-\gamma} \Rightarrow \frac{dP_{\text{total}}}{dV} = -\gamma \cdot \text{const}_2 V^{-\gamma-1} \\ &= -\gamma \cdot \text{const}_2 V^{-\gamma} \frac{1}{V} \Rightarrow \frac{dP_{\text{total}}}{P_{\text{total}}} = -\gamma \frac{dV}{V} \end{aligned}$$

and here we have our relationship between relative changes in P_{total} and V . Now, $dP_{\text{total}} = p$, the sound pressure, and $P_{\text{total}} \approx P_{\text{atm}}$ if $|p| \ll P_{\text{atm}}$. Also, we saw under method 1 that in 1D, $dV/V = d\xi/dx$, so

$$p = -\gamma P_{\text{atm}} \frac{d\xi}{dx}$$

If we compare with the expression in Eq. (3.15), we see that the difference is the factor $\gamma = 1.43$ for air. As a last step, we take the derivative with respect to time, so that we can replace ξ with v_x :

$$\frac{\partial p}{\partial t} = -\gamma P_{\text{atm}} \frac{\partial^2 \xi}{\partial x \partial t}$$

and if we assume that ξ is a smooth function as function of both x and t , we can replace $\partial \xi / \partial t$ by v_x :

$$\frac{\partial p}{\partial t} = -\gamma P_{\text{atm}} \frac{\partial v_x}{\partial x} \quad (3.17)$$

Now, after analyzing the air’s mass behaviour and its spring behaviour, we can take $\frac{\partial}{\partial x}$ of Eq. (3.12), and $\frac{\partial}{\partial t}$ of Eq. (3.17), and we get two versions of $\frac{\partial^2 v_x}{\partial x \partial t}$ (if we assume that the order of $\partial x \partial t$ can be switched), so we can set that they must be equal:

$$-\frac{1}{\rho_0} \frac{\partial^2 p}{\partial x^2} = -\frac{1}{\gamma P_{\text{atm}}} \frac{\partial^2 p}{\partial t^2}$$

and this gives:

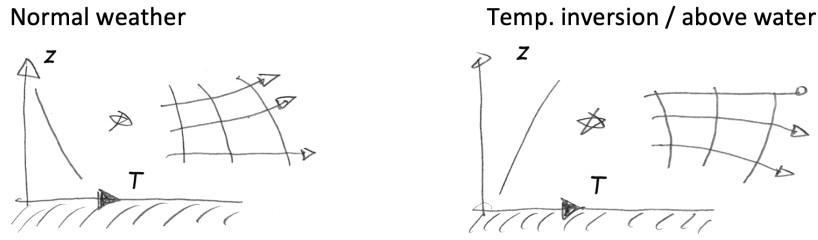


Figure 3.10: A temperature gradient, as function of height over the ground, leads to a variation in the speed of sound with height. Such a variation, in turn, leads to that sound waves bend upwards (left) or downwards (right).

The wave equation in 1D, which describes the sound pressure as function of space and time, $p(x, t)$. We introduce a constant c , which has the unit of m/s:

$$\frac{\partial^2 p}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} = 0 \quad (3.18)$$

where

$$c = \sqrt{\frac{\gamma P_{\text{atm}}}{\rho_0}}$$

and this c is the wave propagation speed.

We will see what the solution to this PDE looks like, but we can already reveal that this wave equation describes non-dispersive waves which propagate in the positive and negative x -direction.

The expression for the speed of sound can be analyzed further, and it is possible to derive a dependence on temperature which will approximately be

$$c \approx (331.6 + 0.6 \cdot T_{\text{centigrade}}) \text{ m/s}$$

where $T_{\text{centigrade}}$ is the temperature in degrees centigrade. This dependence on the temperature, in turn, leads to interesting effects for sound waves propagating outdoors. Normally, the temperature is higher at the ground and lower further up in the atmosphere. Such a temperature gradient leads to that sound waves “bend upwards”, away from the ground, which is good for noise control. Sometimes, however, there is a temperature inversion, and sound waves bend downwards towards the ground, and sound/noise can reach unexpectedly far, see Fig. 3.10.

3.3 The harmonic version of the wave equation - Helmholtz' equation

In Section 3.1, we used the solution without discussing how that solution was derived, and there, we studied the harmonic case where the time-dependence

had the form $e^{j\omega t}$. If we thus *assume that the sound pressure has this form* (and we accept the use of the symbol p both for the function of x , and for the function of x and t):

$$p(x, t) = p(x)e^{j\omega t}$$

and insert in the wave equation, Eq. (3.18), we get

$$\frac{\partial^2 [p(x)e^{j\omega t}]}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 [p(x)e^{j\omega t}]}{\partial t^2} = 0$$

Here we can use the great advantage of the $j\omega$ -method: after carrying out the time-derivative, we can remove the time-harmonic factor, $e^{j\omega t}$, which appears in both terms. Then we don't have a *partial* DE anymore, but an *ordinary* DE,

$$\frac{d^2 p(x)}{dx^2} + \frac{\omega^2}{c^2} p(x) = 0 \Leftrightarrow \frac{d^2 p(x)}{dx^2} + k^2 p(x) = 0$$

This time-harmonic special case of the wave equation has got its own name: *Helmholtz equation*, which in 1D is

$$\frac{d^2 p}{dx^2} + k^2 p = 0 \quad (3.19)$$

Helmholtz equation is an ordinary differential equation of second order, and if we look back at math courses, we might have seen that a DE of the type in Eq. (3.19) has the solution

$$p(x) = a \cdot e^{bx}$$

and we expect that b can have two possible solutions for a second order DE. Inserting this solution form into Eq. (3.19) leads to (a and e^{bx} both vanish)

$$b^2 + k^2 = 0 \Rightarrow b = \pm jk$$

that is,

$$p(x) = e^{\pm jkx}$$

which are exactly the two forms we looked at earlier, for a wave travelling in the positive x -direction and negative x -direction, respectively.

Chapter 4

Acoustic quantities

4.1 The complementary quantity - the particle velocity

We need a complementary quantity to the sound pressure (in the same way as we need voltage and current in electrical circuits), and in acoustics that quantity is the *particle velocity*, $\mathbf{v}(\mathbf{x}, t)$, which is a vector quantity, whereas the sound pressure is a scalar quantity, as pointed out earlier. A vector quantity means that the particle velocity has a *value and a direction*, or, equivalently, that the particle velocity has x -, y -, and z -components, as shown in Eq. (4.1) below and illustrated in Fig. 4.1.

We saw that a plane wave in the positive x -direction has a sound pressure which is described by $p(x, t) = Ae^{j(\omega t - kx)}$, but how about the particle velocity?

First, if a plane wave (PW) is propagating in the x -direction only, then all movements of air particles are also in the x -direction only. So, the vector quantity, $\mathbf{v}$, has an x -component only:

$$\mathbf{v} = v_x \mathbf{e}_x + v_y \mathbf{e}_y + v_z \mathbf{e}_z \Rightarrow \mathbf{v}_{PW,x} = v_x \mathbf{e}_x \quad (4.1)$$

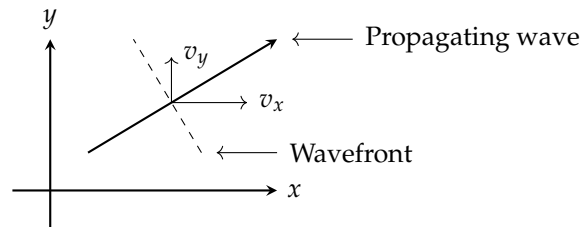


Figure 4.1: Illustration of the particle velocity, $\mathbf{v} = v_x \mathbf{e}_x + v_y \mathbf{e}_y$, in a sound field. The particle velocity has both amplitude and direction in every point, whereas the sound pressure has amplitude only.

where $\mathbf{e}_n$ is the unit vector in the n -direction.

Second, the component, v_x , which can generally be a function of x, y, z : $v_x(\mathbf{x}, t)$, is also a function of x only:

$$v_x(\mathbf{x}, t)_{PW,x} = v_x(x, t)$$

We saw that Newton's 2nd law lead to Eq. (3.12): $\frac{\partial p}{\partial x} = -\rho_0 \frac{\partial v_x}{\partial t}$. Since we are using the complex notation, and harmonic signals, all sound field quantities have the common factor $e^{j\omega t}$:

$$v_x(x, t) = v(x) e^{j\omega t}$$

and therefore,

$$\frac{\partial v_x(x, t)}{\partial t} = j\omega v_x(x)$$

If we then insert this expression, and the known solution for a plane wave in the positive x -direction, $p_{PW,x}^+$, which was given in Eq. (3.3), into Eq. (3.12), we get

$$\begin{aligned} \frac{d(Ae^{-jkx})}{dx} &= -j\omega\rho_0 v_{x,PW}^+(x) \Rightarrow -jkAe^{-jkx} = -j\omega\rho_0 v_{x,PW}^+(x) \\ \Rightarrow v_{x,PW}^+(x) &= \frac{Ae^{-jkx}}{\rho_0 c} = \frac{p_{PW}^+(x)}{\rho_0 c} \end{aligned} \quad (4.2)$$

For a plane wave in the negative x -direction, we can go through the same exercise and find that

$$\Rightarrow v_{x,PW}^-(x) = -\frac{Be^{jkx}}{\rho_0 c} = -\frac{p_{PW}^-}{\rho_0 c} \quad (4.3)$$

As we have stated earlier, the sound pressure, p , is our primary quantity of interest, but often boundary conditions specify the particle velocity. A very common example: if we have a vibrating surface, which could be a loudspeaker element, with a known vibration excursion and therefore also a known vibration velocity of the surface, $v_{\text{surface}}(t)$, then we know the following very important relationship:

The particle velocity in the sound field immediately close to a vibrating surface must be identical to the surface vibration velocity.

Equivalently, the particle *displacement* in the sound field, ξ , immediately close to the vibrating surface must be identical to the vibration *displacement* of the surface.

By the way, what is the “particle velocity”? Is it the velocity of the individual air molecules, as they are traveling around in the gas? Well, sort of - it is the *average* of the velocities of a large number of molecules! If we have a box, with no sound source inside, the particle velocity (and the sound pressure) will be zero, but the air molecules are moving around randomly,

and if the temperature increases, they move faster. They collide with each other and with the walls of the box, and change directions. If we could insert a little measurement device that could record all the molecules passing a point, or rather through a little imagined sphere, and we could magically record all the velocities, with directions, of the molecules passing through that sphere – then the average in the x -, y - and z -direction will indeed be zero, if there is no sound inside the box.

But, if one wall of the box started to vibrate sinusoidally, then the air molecules will be reflected off that vibrating wall at points in space and time which will be similar for a large number of molecules. If one then measures all the velocities, the average will suddenly start to show a systematic sinusoidal variation, and that average will be exactly what we call the particle velocity.

Earlier, it was mentioned that the sound pressure signal can be measured with a microphone (because the vibration velocity of the microphone membrane will be proportional to the sound pressure at the microphone). How can we measure the particle velocity? There are actually much fewer sensor types that can measure particle velocity in a sound field. It might be tempting to think that the microphone membrane's vibration velocity must be the same as the particle velocity in the sound field, especially since it was bombastically claimed above that *The particle velocity in the sound field immediately close to a vibrating surface must be identical to the surface vibration velocity*. We have to jump a little bit ahead to understand the subtle, but important difference between an incoming sound wave, towards a microphone, and a radiated sound wave, from a vibrating structure. We try to explain it here, nevertheless.

- A vibrating structure generates an outgoing sound wave. Again: the particle velocity of that outgoing wave must, immediately close to the vibrating surface, be identical to the vibration velocity of the surface.
- For a microphone, there is an incoming sound wave which hits the microphone, and a reflected sound wave is generated when the incoming wave hits the microphone. Therefore, *at the microphone membrane, the particle velocity of the **sum** of the two waves, must be identical to the vibration velocity of the surface of the microphone membrane*. Thus, the microphone membrane's vibration velocity is not the same as the particle velocity of the incoming sound wave in isolation.

Quite a common technique to estimate the particle velocity is to measure it indirectly via Eq. (3.12):

$$\frac{\partial v_x}{\partial t} = -\frac{1}{\rho_0} \frac{\partial p}{\partial x} \approx -\frac{1}{\rho_0} \frac{\Delta p}{\Delta x}$$

That is, if we measure the sound pressure with two microphones that are spaced a distance Δx apart, the difference of the two sound pressure time signals, divided by $-\rho_0 \Delta x$ will give us the particle acceleration $a_x = \frac{\partial v_x}{\partial t}$ and if

we integrate the time signal, we get the v_x -signal. For time-harmonic signals, the time integration is simply a division by $j\omega$.

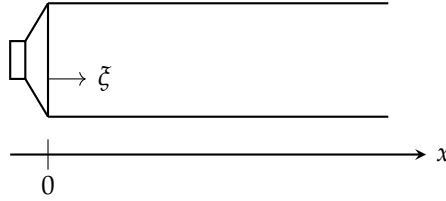


Figure 4.2: Illustration of a loudspeaker in a duct, for Example 1.

Example 1: Plane sound wave

1. A loudspeaker is mounted in a circular tube such that the loudspeaker generates plane waves into the tube, propagating in the positive x -direction, see Fig. 4.2. The loudspeaker is mounted at $x = 0$ and has the same diameter as the tube. The movement of the loudspeaker membrane, $\xi(t)$, is given by

$$\xi(t) = \hat{\xi} \cdot \sin(\omega t)$$

where the amplitude $\hat{\xi}$ is 1 mm, and the frequency f is 50 Hz.

Write an expression for the particle velocity, $v_x(t)$, in the sound wave *immediately in front of* the loudspeaker, at $x = 0$. Use the real form, that is, with sines and/or cosines rather than using the complex exponential factor $e^{j\omega t}$.

Hint: the vibration velocity of the loudspeaker membrane, $v_{\text{ls}}(t)$, is identical to the particle velocity of the sound wave at $x = 0$. Furthermore, the vibration velocity, $v_{\text{ls}}(t)$, is the time derivative of the movement/excursion, $\xi(t)$.

Answer: The vibration velocity is, as pointed out in the hint, the time derivative of the loudspeaker movement/excursion:

$$v_{\text{ls}}(t) = \omega \hat{\xi} \cdot \cos(\omega t)$$

This is also the expression for the particle velocity of the sound wave at $x = 0$, that is, $v_{x, \text{real form}}(0, t) = \omega \hat{\xi} \cdot \cos(\omega t) = \hat{v}_x \cos(\omega t)$. Numerically, $\hat{v}_x = 2\pi \cdot 50 \cdot 0,001 \text{ m/s} = 0,31 \text{ m/s}$.

2. Write an expression for the sound pressure as function of position and time, $p(x, t)$, for the sound wave which propagates away from the loudspeaker. Use the complex form, that is, with the complex exponential factor $e^{j\omega t}$. Air has the density $\rho = 1.2 \text{ kg/m}^3$ and the speed of sound 343 m/s .

Answer: The amplitude of the sound pressure at $x = 0$ is given by the

expression for the (amplitude of the) particle velocity at $x = 0$, from (a), and the acoustic impedance (as given in the list of formulas): $\hat{p} = Z_s \hat{v}_x = \rho_0 c \hat{v}_x$. We first convert the expression for $v_{x,real\ form}(0, t)$ to the complex form,

$$v_{x,real\ form}(0, t) = \hat{v}_x \cos(\omega t) = \text{Re} [v_x(0, t)] = \text{Re} [\hat{v}_x e^{j\omega t}]$$

The sound pressure, on complex form, at $x = 0$ is $p(0, t) = \hat{p} e^{j\omega t}$, and we get the expression for the propagating wave, in the positive x -direction, by multiplying with the propagation factor e^{-jkx} :

$$p(x, t) = \hat{p} e^{-jkx} e^{j\omega t} = \hat{p} e^{j(\omega t - kx)}$$

where $\hat{p} = \rho_0 c \omega \hat{\xi}$ which, numerically, is $\hat{p} = 1,2 \cdot 343 \cdot 2\pi 50 \cdot 0,001 \text{ Pa} = 129 \text{ Pa}$.

Extra As a little extension beyond the scope of this course, it can be mentioned that the two sound field quantities, the scalar quantity p and the vector quantity $\mathbf{v}$, can both be derived from a single sound field quantity called the velocity potential, Φ . It is a scalar quantity which is also governed by the wave equation, like p is. Importantly, one can get p and $\mathbf{v}$ from Φ :

$$p = -\rho \frac{\partial \Phi}{\partial t}$$

$$\mathbf{v} = \nabla \Phi \Rightarrow v_x = \frac{\partial \Phi}{\partial x} \text{ etc}$$

Extra As yet an extension, which you will encounter in the part on EM waves, it can be mentioned that a plane wave, e.g., in the x -direction, assumes that p and v_x are independent of y and z . In many situations this can not be assumed, for instance if a vibrating surface in $x = 0$, as in Example 1 above, does not cover the entire duct cross-section. In more general cases, we have to allow variations of p across the yz -plane. This is done by so-called waveguide modeling, so that a propagating wave in the positive x -direction is modeled as

$$p(x, y, z, t) = A_0 e^{-jkx} + \sum_{\substack{m,n=0 \\ m+n>0}}^{\infty} \varphi_{mn}(y, z) e^{-jk_{x,mn}x} \quad (4.4)$$

where $\varphi_{mn}(y, z)$ are so-called mode functions that describe the variation in the yz -direction. If the duct walls are rigid, which is quite a common case, and the duct has the size l_y by l_z , the mode functions have this form,

$$\varphi_{mn}(y, z) = \cos \frac{m\pi y}{l_y} \cos \frac{n\pi z}{l_z}$$

$k_{x,mn}$ is the wavenumber in the x -direction, which is

$$k_{x,mn} = \sqrt{k^2 - k_{y,m}^2 - k_{z,n}^2} = \sqrt{k^2 - \left(\frac{m\pi}{l_x}\right)^2 - \left(\frac{n\pi}{l_y}\right)^2} \quad (4.5)$$

We can see in the first expression, Eq. (4.4), that the plane wave term is still used, but the variation as function of y and z is described by the infinite sum. This sum is equivalent to that any periodic time signal can be described by a Fourier series, that is, be written as a sum of harmonic frequency components. A periodic variation across space can be written as a sum of harmonically related wavenumber components. The wavenumber $k_{x,mn}$ describes how rapidly, as function of x , that p oscillates down the duct. Eq. (4.5) leads to that higher-order wavenumbers will be imaginary:

$$k_{x,mn} = -jk_{x,mn,\text{imag.}}$$

where $k_{x,mn,\text{imag.}}$ is a positive, real value. (One has to choose a negative value of $k_{x,mn}$ as we will see next.) An imaginary wavenumber implies that the wave is decaying to zero as the wave is propagating:

$$p \sim e^{-jk_x x} = e^{-j(-jk_{x,\text{imag.}})x} = e^{-k_{x,\text{imag.}}x} \xrightarrow{x \rightarrow \infty} 0$$

Maybe this makes it clear why we had to choose the negative value for the square root of a negative number: if we had chosen the positive value, then p would have grown exponentially for increasing x !

The higher the mode number, m and n , the faster will the corresponding mode die out. These modes are called *evanescent modes*, and modes with a real-valued k_x are called *propagating modes*.

4.2 The Doppler effect

In the discussions so far, and further on, we will assume that a sound source, as well as an observer, are both at rest. But, quite a common situation is that the source and/or the receiver is moving towards or away from the receiver/source.



Figure 4.3: The geometrical parameters for the Doppler effect caused by a moving source/and or receiver.

Let's assume that a source starts at position $x_{S,0}$ but moves in the positive x -direction with speed v_S , see Fig. 4.3. Let's also assume that the source emits

pulses every Δt_{emit} which means that the starting position of pulse number n is

$$x_{\text{start},n} = x_{S,0} + v_S t_{\text{emit},n} = x_{S,0} + v_S n \Delta t_{\text{emit}}$$

The pulse propagates with the speed of sound, c , in the positive x -direction, so the position of pulse n at time t is

$$x_n = x_{\text{start},n} + c(t - t_{\text{emit},n}) = x_{S,0} + v_S n \Delta t_{\text{emit}} + c(t - n \Delta t_{\text{emit}})$$

When does pulse n reach the receiver? If the receiver is moving with speed v_R in the positive x -direction, then the receiver position at time t is

$$x_R = x_{R,0} + v_R t$$

So, pulse n reaches the receiver at time $t_{\text{receive},n}$, which must fulfill that $x_n = x_R$:

$$\begin{aligned} x_{S,0} + v_S n \Delta t_{\text{emit}} + c(t_{\text{receive},n} - n \Delta t_{\text{emit}}) &= x_{R,0} + v_R t_{\text{receive},n} \\ \Rightarrow t_{\text{receive},n} &= \frac{x_{R,0} - x_{S,0}}{c - v_R} + \frac{c - v_S}{c - v_R} n \Delta t_{\text{emit}} \end{aligned} \quad (4.6)$$

This means that the receiver receives pulses with a constant interval

$$\Delta t_{\text{receive}} = t_{\text{receive},n+1} - t_{\text{receive},n} = \frac{c - v_S}{c - v_R} \Delta t_{\text{emit}}$$

A pulse train has a spectrum with a fundamental frequency of $1/\Delta t$ and harmonic overtones. Therefore the fundamental frequency at the receiver is

$$f_{\text{receiver}} = f_{\text{sender}} \frac{c - v_R}{c - v_S} \quad (4.7)$$

We can observe that if v_S is larger than v_R (taking the sign of the velocity into account, which means, e.g., that $+0.5$ m/s is larger than -2 m/s), then the distance between source and receiver is shrinking as time goes by, and the received frequency is higher than the emitted one. If v_S is smaller than v_R , the distance between source and receiver is getting larger as time goes by, and the received frequency is lower than the emitted one.

Extra One might wonder what happens if $v_S = c$: Eq. (4.6) makes it seem like all the pulses arrive at the same time to the receiver?!? This is actually what happens, and it results in very strong sound at the receiver: a sonic boom! This happens regularly for airplanes that travel with supersonic speeds, that is, with $v_S > c$, and a cone with the airplane at the peak is following the airplane. The surface area of the cone is a surface with a very high sound pressure, a so-called shock wave, and when this cone surface hits the ground there will be strong disturbance and potential damage due to the very high airpressures.

Often when a physical quantity's amplitude gets very high, the simplified linear mathematical model might predict an infinitely high amplitude. The true system's non-linearities will determine exactly what happens at these high amplitudes.

If the receiver is moving with the speed of sound: $v_R = c$: Eq. (4.6) suggests that the arrival times at the receiver are infinitely high - and it makes sense that the pulses never reach the receiver if the receiver is moving away with the same speed as the sound waves travel.

4.3 Acoustic impedance, Z

In acoustics, there are different quantities called *impedances*, Z . The one we encounter in this course is called *Specific acoustic impedance* and is denoted Z_S , defined as

$$Z_S = \frac{p}{v} \quad (4.8)$$

where p and v must both be quantities with a $j\omega$ -factor removed, that is, the impedance can be used only with harmonic signals, and not signals with general time-dependence. Sometimes, this impedance is just called *acoustic impedance*. Since the impedance involves the particle velocity which is a vector quantity, also the impedance has different components, for the different particle velocity components. Compare this impedance with the electrical impedance which relates the two different quantities in a similar way: $Z_E = e/i$. As we saw above, for a plane wave, we have

$$Z_{S,\text{plane-wave}+} = \rho_0 c, \quad Z_{S,\text{plane-wave}-} = -\rho_0 c \quad (4.9)$$

for the plane wave in the positive and negative x -direction, respectively. Apparently, Z_S depends on which kind of sound field we have (we will see more varieties when we come to spherical waves), which means that even if we know the sound pressure, p , in one point, *we can not from that derive the particle velocity without knowing what kind of sound field we have*. Alternatively, if we know the sound pressure in two points close to each other, we can estimate (one component of) the particle velocity from those two sound pressure signals.

Closely related to the Specific acoustic impedance is the so-called *characteristic impedance*, Z_C , which is a material constant:

$$Z_C = \rho_0 c$$

It might seem confusing that the Specific acoustic impedance is identical with this characteristic impedance, but this is only for the special case of a plane wave (in the positive x -direction).

4.4 Acoustic intensity, I

Acoustic intensity, which has the unit W/m^2 , is given by the time-average of the product of $p_{\text{real}}(t)$ and $\mathbf{v}_{\text{real}}(t)$:

$$\mathbf{I} = \frac{1}{T} \int_0^T p_{\text{real}}(t) \mathbf{v}_{\text{real}}(t) dt$$

We can see what this expression gives for a plane wave:

$$\begin{aligned} \begin{cases} p_{\text{real}}(x, t) = A \cos(\omega t - kx) \\ v_{x, \text{real}}(x, t) = \frac{A}{\rho_0 c} \cos(\omega t - kx) \end{cases} &\Rightarrow p_{\text{real}}(t) v_{x, \text{real}}(t) = \frac{A^2}{\rho_0 c} \cos^2(\omega t - kx) \\ &= \frac{A^2}{\rho_0 c} \frac{1}{2} [1 + \cos 2(\omega t - kx)] \end{aligned}$$

The (x -component of the) intensity will be given by the following integral,

$$I_x = \frac{1}{T} \int_0^T p_{\text{real}}(t) \mathbf{v}_{x, \text{real}}(t) dt$$

where the integration time T is the same as the period of the single-frequency signal. The term $\cos 2(\omega t - kx)$ will vanish when we integrate over such a time period, so what remains is

$$I_x = \frac{1}{T} \int_0^T \frac{A^2}{\rho_0 c} \frac{1}{2} dt = \frac{A^2}{2\rho_0 c} = \frac{A^2}{2\rho_0 c} = \frac{p_{\text{in}, \text{rms}}^2}{\rho_0 c} \quad (4.10)$$

As a little extension, we take a brief look at the sound power quantity. We can acknowledge that if we know the intensity, then we can also find out how much power, W , that passes through a certain surface S_{enclosed} :

$$W = \int_{S_{\text{enclosed}}} \mathbf{I} \cdot \mathbf{n} dS$$

where $\mathbf{I} \cdot \mathbf{n}$ is the component of $\mathbf{I}$ which is perpendicular/normal to the surface. For a plane wave, this is easy to solve for a surface S_{enclosed} which is perpendicular to the propagation direction, x , since I is constant:

$$W = I_x \cdot S_{\text{integration}} = \frac{p_{\text{in}, \text{rms}}^2}{\rho_0 c} S_{\text{integration}}$$

We can also check that the unit, watt, is correct.*

4.5 Sound pressure levels

It is common to express the amplitude of a sound wave in terms of the *sound pressure level*, L_p , expressed in dB, relative to a reference value of 20 μPa , defined as

$$L_p = 10 \log \frac{p_{\text{rms}}^2}{p_{\text{ref}}^2} = 20 \log \frac{p_{\text{rms}}}{p_{\text{ref}}} \quad (4.11)$$

where p_{rms} is the rms value of the sound pressure signal: the square root of the average of the sound pressure squared

$$p_{\text{rms}}^2 = \frac{1}{T} \int_0^T p^2(t) dt$$

*In acoustics, the symbol W is used for power instead of the more common P in electronics. The reason is that P is easy to confuse with the p for sound pressure.

which for a harmonic (single-frequency) time signal is $p_{rms, \text{single-frequency}} = |p| / \sqrt{2}$. Some examples of sound pressure levels are given in Table 1.

From the values in Table 1, we can perhaps realize that the reference value for sound pressure has been chosen such that the just audible level should be around 0 dB. By comparing the given value for p_{rms} to the static air pressure, P_{atm} , normally around 1030 mbar = 103 kPa, we realize that for most sound signals, p is indeed $\ll P_{\text{atm}}$. Also the other acoustical quantities can be expressed as levels, and the most common one is the sound power level, L_W :

$$L_W = 10 \log \frac{W}{W_{ref}} \quad (4.12)$$

where W is the sound power radiated by a sound source, and $W_{ref} = 1 \text{ pW}$ (yes, the power carried by a sound wave is often remarkably small!).

Table 4.1: Some examples of sound pressure levels

L_p [dB] re. 20 μPa	p_{rms} [Pa]	Example
140	200 Pa	Close to a jet engine (3m dist.)
130	63 Pa	Sound causes pain
120	20 Pa	Loud rock concert
110	6.3 Pa	Accelerating motorcycle (5m dist.)
100	2 Pa	Jackhammer (2m dist.)
90	0.63 Pa	Inside a noisy factory
80	200 mPa	Noisy vacuum cleaner
70	63 mPa	Near a road with heavy traffic
60	20 mPa	Normal conversation level
50	6.3 mPa	Quiet restaurant
40	2 mPa	Residential area, at night
30	0.63 mPa	Empty cinema
20	0.2 mPa	Leaves rattling
10	63 μPa	Quiet breathing (3m distance)
0	20 μPa	Normal hearing threshold
-10	6.3 μPa	Hearing threshold at some frequency

Chapter 5

Reflection and transmission of sound waves

5.1 Acoustic boundary conditions

To find out what happens to a sound wave at a boundary between two different media the following conditions must be fulfilled:

1. p must be continuous across the boundary, that is, equal at both sides of the boundary.
2. v_n must also be continuous across the boundary, that is, equal at both sides of the boundary[†].
3. If the two media are viscous, then $v_{\text{tangential}}$ must also be continuous (only relevant for oblique incidence). The viscosity of air is so low that it is ignored in the wave equation.

5.2 Reflection and transmission of plane sound waves - perpendicular incidence

We study a case with two media, with characteristic impedances Z_1 and Z_2 . The boundary between the two is at $x = 0$, and we have an incident plane wave in medium 1 towards the boundary. This boundary will cause a reflected and a transmitted wave, as illustrated in Fig. 5.1, and we can set up the general expressions for these three waves, and try to derive the relationship between their amplitudes by using the BCs in Section 5.1. In doing so, we leave out the

[†]For oblique incidence, conditions 1 and 2 imply what is known as "Snell's law", see below.

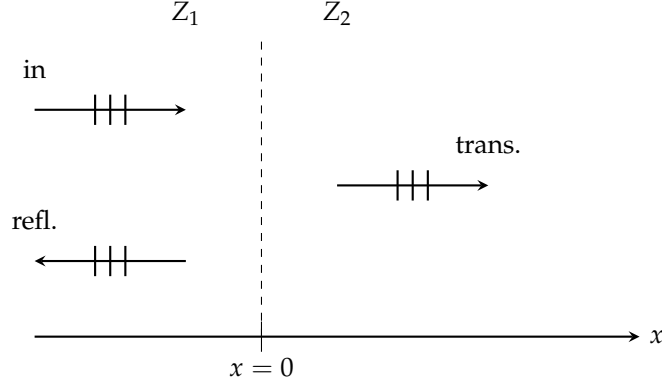


Figure 5.1: Two media, with characteristic impedances Z_1 and Z_2 .

common time-harmonic factor $e^{j\omega t}$. Thus, in medium 1, $x < 0$, we have two waves: the incident and reflected ones:

$$\begin{cases} p_{\text{in}}(x) = Ae^{-jk_1x}, & v_{x,\text{in}}(x) = \frac{A}{Z_1}e^{-jk_1x} \\ p_{\text{refl}}(x) = Be^{jk_1x}, & v_{x,\text{refl}}(x) = -\frac{B}{Z_1}e^{jk_1x} \end{cases}$$

where $k_1 = \omega/c_1$, if c_1 is the wave propagation speed in medium 1. In medium 2, $x > 0$, no reflected wave is generated, because there is no change in impedance which could cause a reflection, and no source of a wave either. Thus we have a transmitted wave only:

$$p_{\text{trans}}(x) = Ce^{-jk_2x}, \quad v_{x,\text{trans}}(x) = \frac{C}{Z_2}e^{-jk_2x}$$

with $k_2 = \omega/c_2$ where c_2 is the wave propagation speed in medium 2. We apply BC 1 and 2 at $x = 0$:

$$p(0^-) = p(0^+) \Rightarrow A + B = C$$

$$v_x(0^-) = v_x(0^+) \Rightarrow \frac{A - B}{Z_1} = \frac{C}{Z_2}$$

From these two equations we can derive:

The reflection and transmission factors for the sound pressure:

$$R_p \equiv \frac{p_{\text{refl}}(0^-)}{p_{\text{in}}(0^-)} = \frac{B}{A} = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad (5.1)$$

$$T_p \equiv \frac{p_{\text{trans}}(0^+)}{p_{\text{in}}(0^-)} = \frac{C}{A} = \frac{2Z_2}{Z_2 + Z_1} \quad (5.2)$$

These two factors are related as

$$T_p = R_p + 1 \quad (5.3)$$

Some interesting special cases are:

- $R_p = 0$ (perfect absorption)

This implies no reflection at all, which would happen for a perfect sound absorber. To obtain this, we need that $Z_2 = Z_1$. Eq. (5.3) also tells us that $T_p = 1$, which seems to make sense: if there is no reflection, then the transmitted wave will have exactly the same amplitude as the incident wave.

- $R_p = 1$ (perfect reflection)

This implies a perfect reflection, and such a boundary is sometimes called a "rigid" wall. To obtain this, we need that $Z_2 \gg Z_1$. Now Eq. (5.3) gives us a result which seems weird: $T_p = 2$, which suggests that the transmitted wave is stronger than the reflected wave?? To understand this, we can inspect a case where R_p is *almost* 1. We choose $Z_2 = 1000 Z_1$, which gives $R_p \approx 0.998$ and $T_p \approx 1.998$. The transmitted wave does indeed appear to be stronger than the reflected wave. If we, however, inspect the *intensities* of the reflected and transmitted waves, it makes more sense. Thus, using Eq. (4.10) we can express the reflected and transmitted intensities as

$$I_{\text{refl}} = \frac{p_{\text{refl,rms}}^2}{Z_1} = p_{\text{in,rms}}^2 \frac{|R_p|^2}{Z_1}, \quad I_{\text{trans}} = \frac{p_{\text{trans,rms}}^2}{Z_2} = p_{\text{in,rms}}^2 \frac{|T_p|^2}{Z_2}$$

$$\Rightarrow \frac{I_{\text{trans}}}{I_{\text{refl}}} = \frac{|T_p|^2}{|R_p|^2} \frac{Z_1}{Z_2}$$

For our numerical example, we get

$$\frac{I_{\text{trans}}}{I_{\text{refl}}} \approx \frac{1.998^2}{0.998^2} \frac{1}{1000} \approx 0,004$$

So, even if the transmitted pressure has a higher amplitude than the reflected pressure, the transmitted intensity is much weaker than the reflected intensity*.

- $R_p = -1$ (perfect, phase-shifted reflection)

This requires that $Z_2 \ll Z_1$, and a common situation for this is when medium 1 is water and medium 2 is air. For this special case, we apparently get $T_p = 0$, that is, no transmitted wave whatsoever.

*An additional effect, which goes beyond the scope of this course, is the fact that for a plane wave with *oblique* incidence (a case described in section 5.4), the transmitted wave will die out quickly inside medium 2. This decay happens because the wave number of the transmitted wave, k_2 , will become *complex*, which leads to that the plane wave dies out, rather than propagate for ever.

Example 2: Reflection of plane wave

An important application of the plane wave case is the measurement of the reflection coefficient, R_p , for a material sample. A material with unknown impedance Z_2 can be placed inside a tube, see Fig. 5.2. The sound field inside the tube (for $x < 0$), can be written as an incident sound wave and a reflected one (again, we remove the shared factor $e^{j\omega t}$ from all terms. In addition, we write the wave-number as k rather than k_1 because in this example we don't really care about what happens in medium 2):

$$p_{\text{total}}(x) = p_{\text{in}}(x) + p_{\text{refl}}(x) = A \left[e^{-jkx} + R_p e^{jkx} \right]$$

A classical version of this setup is called "Kundt's tube", where one just measures the magnitude, $|p_{\text{total}}(x)|$, as function of x . By finding the values of the maxima and minima, one can derive the value of R_p . The magnitude of $|p_{\text{total}}(x)|$ is derived as follows:

$$p_{\text{total}}(x) = A e^{-jkx} \left[1 + R_p e^{j2kx} \right] \Rightarrow |p_{\text{total}}(x)| = |A| \left| 1 + R_p e^{j2kx} \right|$$

Here we take the step to write the complex R_p as $R_p = |R_p| e^{j\varphi}$:

$$\begin{aligned} |p_{\text{total}}(x)| &= |A| \left| 1 + |R_p| e^{j(2kx + \varphi)} \right| \\ &= |A| \sqrt{[1 + |R_p| \cos(2kx + \varphi)]^2 + [|R_p| \sin(2kx + \varphi)]^2} \\ &= |A| \sqrt{1 + |R_p|^2 + 2|R_p| \cos(2kx + \varphi)} \end{aligned}$$

Since we know that $\cos y \in [-1, +1]$, we know what the minimum and maximum value of $|p_{\text{total}}(x)|$ can be:

$$\max(|p_{\text{total}}|) = |A| \sqrt{1 + |R_p|^2 + 2|R_p|} = |A| (1 + |R_p|)$$

$$\min(|p_{\text{total}}|) = |A| \sqrt{1 + |R_p|^2 - 2|R_p|} = |A| (1 - |R_p|)$$

and, the uninteresting amplitude A vanishes if we divide the maximum value by the minimum value, to get the so-called *pressure standing-wave ratio*

$$PSWR = \frac{\max(|p_{\text{total}}|)}{\min(|p_{\text{total}}|)} = \frac{1 + |R_p|}{1 - |R_p|}$$

The *PSWR* is easy to measure, and we can apparently find $|R_p|$ if we know *PSWR*. In addition, we can figure out what the unknown phase

shift φ is from the location of the first minimum since $x_{\min}$ is when $\cos(2kx_{\min} + \varphi) = -1$:

$$2kx_{\min} + \varphi = \pm\pi, \pm3\pi, \dots$$

What we observe in our analysis is the so-called *interference effects* between the two sound waves: we can get positive interference, when the total sound pressure is amplified by the superposition, and we can get negative interference, when the total sound pressure is reduced by the superposition.

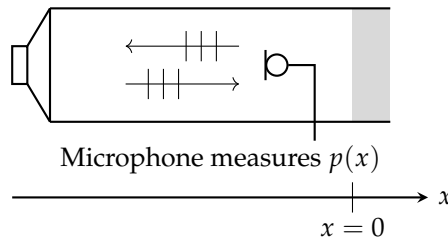


Figure 5.2: A duct with a material sample with an unknown impedance, Z_2 placed at $x = 0$, and a loudspeaker which generates a plane wave towards the material sample.

Example 3: Reflection of a plane wave at a rigid wall

With the same setup as in example 2 above, but with a known R_p , we get the sound field expressions for $x < 0$ (again, with the common factor $e^{j\omega t}$ removed):

$$p_{\text{total}}(x) = p_{\text{in}}(x) + p_{\text{refl}}(x) = A \left[e^{-jkx} + R_p e^{jkx} \right]$$

$$v_{x,\text{total}}(x) = v_{x,\text{in}}(x) + v_{x,\text{refl}}(x) = \frac{A}{Z_1} \left[e^{-jkx} - R_p e^{jkx} \right]$$

Now, a common and important special case is when $R_p = 1$, that is, the wall is perfectly rigid:

$$R_p = 1 \Rightarrow \begin{cases} p_{\text{total}}(x) = A \left[e^{-jkx} + e^{jkx} \right] = A 2 \cos kx \\ v_{x,\text{total}}(x) = \frac{A}{Z_1} \left[e^{-jkx} - e^{jkx} \right] = \frac{A}{Z_1} (-2j) \sin kx \end{cases}$$

We can observe various things from these expressions:

- In the expression for $v_{x,\text{total}}$, we see that v_x will be zero at $x = 0$. This is exactly what happens at a “perfectly rigid wall”: the wall surface

can not move, and therefore, the particle velocity immediately at the wall must be zero as well.

- If we place a recording microphone near a hard wall, the spectrum of the recording will be modified according to the interference effect described by the $2 \cos kx$ -factor: some regularly placed frequencies will get twice the amplitude of the incident sound wave while frequencies halfway between those peak-frequencies will get a strong attenuation (theoretically down to 0 Pa). In practice, recording microphones are usually placed away from hard reflecting surfaces for this reason. On the other hand, some microphones (so-called boundary microphones) are placed as close as possible to a wall, to benefit from the factor of 2 while pushing the negative-interference frequencies above 20 kHz!
- When the sound pressure level is measured in a room, e.g., to check if the background noise level fulfills some specifications, the measurements should be done at a certain distance from the walls, to avoid this doubling of the sound pressure amplitude.

In acoustics, a common task to solve is to reduce the sound pressure amplitude in various situations, and this might be done by using sound absorbing materials, that is, materials with R_p close to 0, in order to reduce the reflected sound waves. This can also be done by active methods: a loudspeaker can be used instead of a sound absorbing material. A microphone is recording the sound pressure in front of the loudspeaker, and some signal processing algorithm is used to derive how the microphone signal should give a loudspeaker signal such that the reflected sound wave is minimized. This is sometimes called *Active noise control*. It might seem much more complex with a microphone, loudspeaker, amplifier, and signal processing equipment instead of some passive sound absorbing material but in certain applications, active noise control can be very attractive. A primary advantage is that the active system can take much less space than sound absorbing material.

5.3 Resonances in tubes

The duct in figure 5.2 exemplifies that sound waves in ducts, or pipes, or tubes, are very common examples where 1D sound waves, as given by the expressions in Example 3, describe the situation quite well.* We repeat the solution for the special case with a **rigid wall** at $x = 0$:

$$R_p = 1 \Rightarrow \begin{cases} p_{\text{total}}(x) = A2 \cos kx \\ v_{x,\text{total}}(x) = \frac{-2jA}{Z_1} \sin kx \end{cases} \quad (5.4)$$

*The plane-wave model is quite accurate up to a frequency where the first so-called cross-mode can occur, and that happens when the largest cross-dimension is half a wavelength.

When we deal with tubes or pipes, the rigid-wall end is often called a **closed end**. We also present the solution for the special case with a so-called **pressure-release wall** at $x = 0$:

$$R_p = -1 \Rightarrow \begin{cases} p_{\text{total}}(x) = 2jA \sin kx \\ v_{x,\text{total}}(x) = \frac{2A}{Z_1} \cos kx \end{cases} \quad (5.5)$$

Without going into the details, we claim that an **open end** of a tube behaves **almost** like a pressure-release wall, at low frequencies.

In both Eqs. (5.4) and (5.5), the scale factor, or amplitude, A , is determined by the amplitude of the source which is generating the sound field, such as the loudspeaker drawn in Fig. 5.2. If we don't place a loudspeaker at the left end of the tube, we could instead construct a closed end, or an open end, at the left end of the tube, at $x = -L$. If we do, we will enforce one of two alternative boundary conditions:

$$\begin{cases} \text{A closed wall at } x = -L : & v_{x,\text{total}}(-L) = 0, \text{ or} \\ \text{a pressure-release wall at } x = -L : & p_{\text{total}}(-L) = 0 \end{cases}$$

We will see what happens to the solutions for those two alternatives, and we will need to inspect four situations:

1. **Closed end at $x = 0$ and closed end at $x = -L$:** (a closed-closed tube)
Eq. (5.4) fulfills the BC at $x = 0$. To fulfill $v_{x,\text{total}}(-L) = 0$, we see in Eq. (5.4) that we have two possibilities:
 - (a) $A = 0$ - that is, it is completely quiet in the pipe/tube. This is often called the **trivial solution** which is uninteresting.
 - (b) If A is not zero, then it must be that

$$\sin(-kL) = 0 \Rightarrow kL = 0, \pi, 2\pi, 3\pi, \dots$$

This is much more interesting, since this expression tells us that only some frequencies, f_n , can exist inside the closed-closed tube: those which fulfill

$$k_n L = n\pi \Rightarrow f_n = \frac{nc}{2L}$$

These frequencies are the so-called "eigenfrequencies", or resonance frequencies, of the closed-closed tube.

2. **Closed end at $x = 0$ and open end at $x = -L$:** (a closed-open tube)
Again, Eq. (5.4) fulfills the BC at $x = 0$, and to fulfill the BC at $x = -L$, the expression for $p_{\text{total}}(x)$ in Eq. (5.4) gives the solution (we skip the trivial solution)

$$\begin{aligned} p_{\text{total}}(-L) = 0 &\Rightarrow \cos(-kL) = 0 \Rightarrow k_n L = (2n+1)\frac{\pi}{2} \\ &\Rightarrow f_n = \frac{(2n+1)c}{4L} \end{aligned}$$

and these frequencies are apparently the eigenfrequencies/resonance frequencies, of the closed-open tube,

3. **Open end at $x = 0$ and closed end at $x = -L$:** (an open-closed tube)
Eq. (5.5) fulfills the BC at $x = 0$. To fulfill the BC at $x = -L$, the expression for $v_{x,\text{total}}(x)$, in Eq. (5.5) leads to that

$$\cos(-kL) = 0 \Rightarrow k_n L = (2n + 1) \frac{\pi}{2} \Rightarrow f_n = \frac{(2n + 1)c}{4L}$$

Here we see that the eigenfrequencies/resonance frequencies, of the open-closed tube are the same as those of the closed-open tube - which, in retrospect seems obvious: the sound-field can not tell the difference between those two tubes.

4. **The fourth and last possibility: Open end at $x = 0$ and open end at $x = -L$:** (an open-open tube)
Eq. (5.5) fulfills the BC at $x = 0$. To fulfill the BC at $x = -L$, the expression for $p_{\text{total}}(x)$, in Eq. (5.5) leads to that

$$\sin(-kL) = 0 \Rightarrow k_n L = n\pi \Rightarrow f_n = \frac{nc}{2L}$$

Apparently, the eigenfrequencies/resonance frequencies of the open-open tube are the same as those of the closed-closed tube.

A few comments can be made about this analysis:

- We started by removing the loudspeaker from this imagined tube - but how can there be sound at all if there is no loudspeaker (or other sound source)? That is a very good question! The term "eigenfrequency" gives a hint: those are the frequencies that can exist "on their own", without any source. Or, to sound less mysterious: **after a sound source has been turned off**. We can, for instance, send in a pulse sound into the tube. After that pulse has been sent by a sound source, there is no sound source in the tube, and the sound field is "on its own". In that situation, only the eigenfrequencies can exist!
- If we actually have a loudspeaker, playing sound constantly into the tube, then the eigenfrequencies will be amplified strongly by the tube, and this phenomenon is called a **resonance**. **Resonances are extremely important phenomena in most dynamic (moving) systems**. Resonances can cause a very strong response, which can be unwanted or wanted:
 - **Wanted resonances:** many music instruments' sound generation mechanism is based on exactly this phenomenon. We can get strong sound through resonances as a kind of mechanical amplification. **A bonus point is that the resonance frequencies we saw for the various tubes above form a harmonic series, which sounds pleasant to us.** Beyond the scope of this course: resonance frequencies of two-

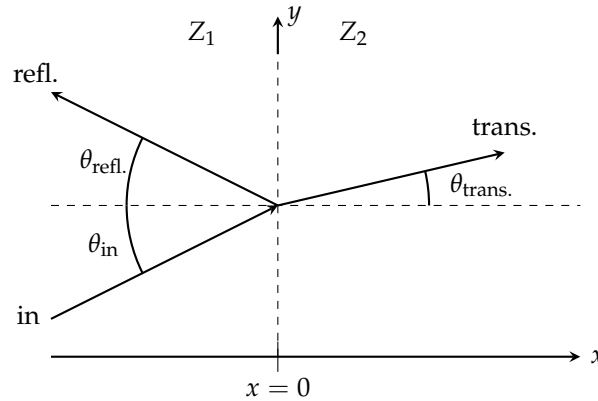


Figure 5.3: An incident wave, in medium 1, hits an interface between two media with an angle θ_{in} , and generates a reflected wave, back into medium 1, and a transmitted wave, into medium 2.

or three-dimensional systems, such as membranes, do not have harmonically distributed resonance frequencies - so the sound from a drum doesn't sound as "tonal" as sounds from flutes, organ pipes, brass and woodwind instruments, string instruments and other instruments which use 1D-resonance systems.

- **Unwanted resonances:** mechanical constructions like buildings, bridges etc can get unwanted resonances that happen to get excited by the ordinary use, or by rare events like very strong winds from one direction etc. Very famous examples include the Tacoma Narrows bridge collapse in 1940, or the London Millenium Footbridge which became "wobbly" at its opening in 2000.

Resonances in tubes can easily be demonstrated: if one takes a cardboard tube, or hard plastic tube, and hits the end with the palm of the hand, you can hear a tonal sound with a low pitch (fundamental frequency = f_1). You can even notice the pitch difference between removing the hand immediately after hitting the tube end, or keeping the hand in place.

5.4 Reflection and transmission of plane sound waves - oblique incidence

Now, if the plane sound wave which hits the interface between media 1 and 2 does not have perpendicular incidence, but hits the interface with an angle θ_{in} ($\theta_{\text{in}}=0$ corresponds to perpendicular incidence), then we have to use the more general expression for the plane sound wave from Eq. (3.7), and also introduce unknown angles θ_{refl} and θ_{trans} , for the reflected and transmitted

wave, respectively, see Fig. 5.3. As before, we suppress the factor $e^{j\omega t}$ and get

$$\begin{aligned} p_{\text{in}}(x, y, t) &= A e^{-jk_1(\cos \theta_{\text{in}})x} e^{-jk_1(\sin \theta_{\text{in}})y} \\ p_{\text{refl}}(x, y, t) &= B e^{jk_1(\cos \theta_{\text{refl}})x} e^{-jk_1(\sin \theta_{\text{refl}})y} \\ p_{\text{trans}}(x, y, t) &= C e^{-jk_2(\cos \theta_{\text{trans}})x} e^{-jk_2(\sin \theta_{\text{trans}})y} \end{aligned}$$

with $k_1 = \omega/c_1$ and $k_2 = \omega/c_2$. Since we are assuming a harmonic time-dependence, the expression for the particle velocity is found using Eq. (3.14),

$$v_x = -\frac{1}{j\omega\rho_1} \frac{\partial p}{\partial x}$$

Thus, we get

$$\begin{aligned} v_{x,\text{in}}(x, y, t) &= -\frac{1}{j\omega\rho_1} (-j)k_1(\cos \theta_{\text{in}}) A e^{-jk_1(\cos \theta_{\text{in}})x} e^{-jk_1(\sin \theta_{\text{in}})y} \\ &= \frac{A \cos \theta_{\text{in}}}{\rho_1 c_1} e^{-jk_1(\cos \theta_{\text{in}})x} e^{-jk_1(\sin \theta_{\text{in}})y} = \frac{A \cos \theta_{\text{in}}}{Z_1} e^{-jk_1(\cos \theta_{\text{in}})x} e^{-jk_1(\sin \theta_{\text{in}})y} \end{aligned}$$

and

$$\begin{aligned} v_{x,\text{refl}}(x, y, t) &= -\frac{B \cos \theta_{\text{refl}}}{Z_1} e^{jk_1(\cos \theta_{\text{refl}})x} e^{-jk_1(\sin \theta_{\text{refl}})y} \\ v_{x,\text{trans}}(x, y, t) &= \frac{C \cos \theta_{\text{trans}}}{Z_2} e^{-jk_2(\cos \theta_{\text{trans}})x} e^{-jk_2(\sin \theta_{\text{trans}})y} \end{aligned}$$

Now we can apply the same continuity (across the interface $x = 0$) conditions as we used for perpendicular incidence. First, continuity for the sound pressure leads to:

$$A e^{-jk_1(\sin \theta_{\text{in}})y} + B e^{-jk_1(\sin \theta_{\text{refl}})y} = C e^{-jk_2(\sin \theta_{\text{trans}})y} \quad (5.6)$$

If this equation should be fulfilled for any y , then all three exponential factors must be identical, so that they can be removed, and we get

$$A + B = C \quad (5.7)$$

The three exponential factors are identical if

$$\begin{aligned} k_1 \sin \theta_{\text{in}} &= k_1 \sin \theta_{\text{refl}} = k_2 \sin \theta_{\text{trans}} \\ \Rightarrow \frac{\omega}{c_1} \sin \theta_{\text{in}} &= \frac{\omega}{c_1} \sin \theta_{\text{refl}} = \frac{\omega}{c_2} \sin \theta_{\text{trans}} \end{aligned}$$

By equating the first two terms, we get the condition for a so-called *specular reflection*,

$$\theta_{\text{refl}} = \theta_{\text{in}} \quad (5.8)$$

Furthermore, by equating the first and the last terms, we get *Snell's law*

$$\frac{\sin \theta_{\text{in}}}{c_1} = \frac{\sin \theta_{\text{trans}}}{c_2} \quad (5.9)$$

This is a central relationship for the transmission of all kinds of waves,

from one medium to another. Snell's law can be written in the most useful form, which tells what the transmission angle will be, if we know the incidence angle, and the two propagation speeds,

$$\sin \theta_{\text{trans}} = \frac{c_2}{c_1} \sin \theta_{\text{in}}$$

But, how can this relationship work - if the fraction c_2/c_1 is large enough, then $\sin \theta_{\text{trans}}$ could be larger than one, and the value of the sine function can never exceed one?? The solution to this is that there will be *no transmitted wave if c_2/c_1 is large enough - there will be a total reflection!* Formulated differently: there will be a maximum incidence angle, a critical angle, θ_c , beyond which there will be total reflection:

$$\frac{c_2}{c_1} \sin \theta_{\text{in}} < 1 \Rightarrow \theta_{\text{in,max}} = \theta_c = \arcsin \frac{c_1}{c_2}$$

One consequence of this is that the interface between the two media will behave as a mirror for light with large enough incidence angles.

Second, continuity for v_x leads to:

$$\frac{A \cos \theta_{\text{in}}}{Z_1} e^{-jk_1(\sin \theta_{\text{in}})y} - \frac{B \cos \theta_{\text{refl}}}{Z_1} e^{-jk_1(\sin \theta_{\text{refl}})y} = \frac{C \cos \theta_{\text{trans}}}{Z_2} e^{-jk_2(\sin \theta_{\text{trans}})y} \quad (5.10)$$

Using the result in Eq. (5.8), we can rewrite Eq. (5.10) as

$$\frac{A \cos \theta_{\text{in}}}{Z_1} e^{-jk_1(\sin \theta_{\text{in}})y} - \frac{B \cos \theta_{\text{in}}}{Z_1} e^{-jk_1(\sin \theta_{\text{in}})y} = \frac{C \cos \theta_{\text{trans}}}{Z_2} e^{-jk_2(\sin \theta_{\text{trans}})y}$$

and again, the exponential factors are identical and can be factored out, so that

$$\frac{A \cos \theta_{\text{in}}}{Z_1} - \frac{B \cos \theta_{\text{in}}}{Z_1} = \frac{C \cos \theta_{\text{trans}}}{Z_2} \quad (5.11)$$

We can solve Eqs. (5.7) and (5.11) by getting rid of C and arrive at an expression for the reflection factor for the sound pressure, R_p , for oblique incidence,

$$R_p = \frac{B}{A} = \frac{Z_2 \cos \theta_{\text{in}} - Z_1 \cos \theta_{\text{trans}}}{Z_2 \cos \theta_{\text{in}} + Z_1 \cos \theta_{\text{trans}}} \quad (5.12)$$

where the angle θ_{trans} can be found from Snell's law in Eq. (5.9). Likewise, we can instead solve Eqs. (5.7) and (5.11) by getting rid of B and arrive at an expression for the transmission factor for the sound pressure, T_p , for oblique incidence,

$$T_p = \frac{C}{A} = \frac{2Z_2 \cos \theta_{\text{in}}}{Z_2 \cos \theta_{\text{in}} + Z_1 \cos \theta_{\text{trans}}} \quad (5.13)$$

Extra In some practical problems in acoustics, medium number 2 has a much lower wave speed than medium 1, and then Snell's law implies that $\theta_{\text{trans}} \approx 0$. Such materials are called **locally reacting** and the expression for the reflection factor is simplified to

$$R_{p,\text{locally reacting}} = \frac{Z_2 \cos \theta_{\text{in}} - Z_1}{Z_2 \cos \theta_{\text{in}} + Z_1}$$

Extra We might get a little curious about the y -component of the particle velocity - don't we also need to check that it is continuous across the interface $x = 0$? Let's check! We can find the expressions for v_y from the y -component version of Eq. (3.14)

$$v_y = -\frac{1}{j\omega\rho_1} \frac{\partial p}{\partial y}$$

so we can derive the expressions for v_y of the incident, reflected and transmitted waves (we use the known fact that $\theta_{\text{refl}} = \theta_{\text{in}}$):

$$v_{y,\text{in}}(x, y, t) = \frac{A \sin \theta_{\text{in}}}{Z_1} e^{-jk_1(\cos \theta_{\text{in}})x} e^{-jk_1(\sin \theta_{\text{in}})y}$$

$$v_{y,\text{refl}}(x, y, t) = \frac{B \sin \theta_{\text{in}}}{Z_1} e^{jk_1(\cos \theta_{\text{in}})x} e^{-jk_1(\sin \theta_{\text{in}})y}$$

$$v_{y,\text{trans}}(x, y, t) = \frac{C \sin \theta_{\text{trans}}}{Z_2} e^{-jk_2(\cos \theta_{\text{trans}})x} e^{-jk_2(\sin \theta_{\text{trans}})y}$$

Then, on the left side of $x = 0$, we get

$$v_y(0^-, y, t) = (A + B) \frac{\sin \theta_{\text{in}}}{Z_1} e^{-jk_1(\sin \theta_{\text{in}})y}$$

and we can replace $A + B$ with C according to Eq. (5.7),

$$v_y(0^-, y, t) = C \frac{\sin \theta_{\text{in}}}{Z_1} e^{-jk_1(\sin \theta_{\text{in}})y} = \frac{1}{\rho_1} C \frac{\sin \theta_{\text{in}}}{c_1} e^{-jk_1(\sin \theta_{\text{in}})y}$$

On the right side of $x = 0$, we get

$$v_y(0^+, y, t) = C \frac{\sin \theta_{\text{trans}}}{Z_2} e^{-jk_2(\sin \theta_{\text{trans}})y} = \frac{1}{\rho_2} C \frac{\sin \theta_{\text{trans}}}{c_2} e^{-jk_2(\sin \theta_{\text{trans}})y}$$

Now, Snell's law leads to that everything after the initial factor $1/\rho_{1,2}$ is identical, so that

$$v_y(0^+, y, t) = \frac{\rho_1}{\rho_2} v_y(0^-, y, t)$$

Apparently, we do not get continuity for v_y across the interface (unless $\rho_2 = \rho_1$). Isn't that a problem?? No! The wave equation that we derived did not involve friction/viscosity in the gas/fluid, so then the particles next to each other,

at $x = 0$, can "slide" against each other in the y -direction! In reality, there will always be some small viscosity, and there will be a thin surface layer where there is a gradual transition of v_y from medium 1 to medium 2. This surface layer does, however, not affect the p and the v_x functions that we analyzed above.

Chapter 6

Waves in 3D - Spherical sound waves

6.1 Waves in 3D: the 3D wave equation

Sound waves can also be studied in 2D and 3D, and in various coordinate systems (cylindrical, ellipsoidal etc). We give, without discussing the details of the derivation, the wave equation in 3D:

$$\nabla^2 p(\mathbf{x}, t) - \frac{1}{c^2} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} = 0 \quad (6.1)$$

where $\nabla^2 p$ is the so-called Laplace operator, which is sometimes written Δp , and takes different forms in different coordinate systems. We look at the two most common ones, the cartesian:

$$\nabla^2 p(x, y, z) = \frac{\partial^2 p(x, y, z)}{\partial x^2} + \frac{\partial^2 p(x, y, z)}{\partial y^2} + \frac{\partial^2 p(x, y, z)}{\partial z^2} \quad (6.2)$$

and the spherical:

$$\nabla^2 p(r, \theta, \varphi) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial p(r, \theta, \varphi)}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial p(r, \theta, \varphi)}{\partial \theta} \right) \quad (6.3)$$

$$+ \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 p(r, \theta, \varphi)}{\partial \varphi^2} \quad (6.4)$$

We realize that if we know that our sound field depends only on x , $p(\mathbf{x}) = p(x)$, then Eqs. (6.1) and (6.2) together give the 1D wave equation, Eq. (3.18). For the spherical coordinate system, which is required to study the radiation from point sources, we will initially look only at rotationally symmetrical waves, that is, $p(\mathbf{x}) = p(r)$, and then Eqs. (6.1) and (6.4) will lead to

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial p(r, t)}{\partial r} \right) - \frac{1}{c^2} \frac{\partial^2 p(r, t)}{\partial t^2} = 0$$

$$\Leftrightarrow \frac{\partial^2 p(r, t)}{\partial r^2} + \frac{2}{r} \frac{\partial p(r, t)}{\partial r} - \frac{1}{c^2} \frac{\partial^2 p(r, t)}{\partial t^2} = 0$$

Now, it turns out that we get exactly the same result if we rewrite as*

$$\frac{\partial^2 (r \cdot p(r, t))}{\partial r^2} - \frac{1}{c^2} \frac{\partial^2 (r \cdot p(r, t))}{\partial t^2} = 0$$

If we compare this PDE with the 1D wave equation, Eq. (3.18), we realize that the quantity $r \cdot p(r)$ must have the same two solutions as $p(x)$, and then

$$\begin{aligned} r \cdot p(r) &= A e^{-jkr} + B e^{jkr} \\ \Rightarrow p(r) &= A \frac{e^{-jkr}}{r} + B \frac{e^{jkr}}{r} \end{aligned} \quad (6.5)$$

and these two terms are the outgoing and incoming spherical waves, respectively.

Extra As a little sidenote, which is beyond the scope of sound waves in this course: if the sound field is a function of r and θ , then we have more complicated solutions, which have this mathematical form for an outgoing wave

$$p(r, \theta) = A_0 \frac{e^{-jkr}}{r} + \sum_{n=1}^{\infty} A_n h_n^{(1)}(kr) P_n(\cos \theta) \quad (6.6)$$

where $h_n^{(1)}$ are so-called spherical Hankel functions of the first kind:

$$h_0^{(1)} = \frac{e^{-jkr}}{-jkr}, \quad h_1^{(1)} = \frac{e^{-jkr}}{(kr)^2} (j - kr), \dots$$

A_n are amplitudes for each term in the sum, and the functions $P_n(\cos \theta)$ are so-called Legendre polynomials:

$$P_0 = 1, \quad P_1 = \cos \theta, \quad P_2 = \frac{1}{2} (2 \cos^2 \theta - 1), \dots$$

We realize that the first term in Eq. (6.6) is the omnidirectional outgoing spherical wave we saw above, in Eq. (6.5). We also realize that the first term in Eq. (6.6) could also have been included in the sum, as the term $n = 0$.

If the sound field is a function of the second angle as well, $p = p(r, \theta, \varphi)$, then a slightly more complicated form than in Eq. (6.6) results.

6.2 The spherical sound wave solution

As we saw above, in Eq. (6.5), the solution for harmonic waves is

$$p(r, t) = \left(A \frac{e^{-jkr}}{r} + B \frac{e^{jkr}}{r} \right) e^{j\omega t} \quad (6.7)$$

*Remember that $\frac{\partial^2 (r \cdot p(r, t))}{\partial r^2}$ is derived as $\frac{\partial}{\partial r} \left[\frac{\partial (r \cdot p(r, t))}{\partial r} \right]$

which is the form for an outgoing spherical wave, from the origin, and an incoming spherical wave. Practically always, we can discard the second solution since it is only in very special geometries that we could assume that such a solution could exist (one possible scenario would be the sound field inside a spherical volume*). So, the useful solution form is

$$p(r) = A \frac{e^{-jkr}}{r} \quad (6.8)$$

The particle velocity expression (for an outgoing spherical wave) can be found from the gradient of the sound pressure expression in Eq. (6.8), which implies

$$\begin{aligned} \nabla p &= \frac{\partial}{\partial r} \left(A \frac{e^{-jkr}}{r} \right) = A \left(-\frac{e^{-jkr}}{r^2} - \frac{jke^{-jkr}}{r} \right) = -A \frac{e^{-jkr}}{r} \left(\frac{1}{r} + jk \right) \\ &= -p(r) \left(\frac{1}{r} + jk \right) \end{aligned}$$

Similarly to Eq. (3.12) for plane waves, the gradient of p , $\nabla p(r)$ should equal $-\rho_0 \frac{\partial v_r(r,t)}{\partial t}$ which, for time-harmonic signals, equals $-j\omega\rho_0 v_r(r)$, so

$$\begin{aligned} -A \frac{e^{-jkr}}{r} \left(\frac{1}{r} + jk \right) &= -j\omega\rho_0 v_r(r) \Rightarrow v_r(r) = \frac{A}{j\omega\rho_0} \frac{e^{-jkr}}{r} \left(\frac{1}{r} + jk \right) \\ \Leftrightarrow v_r(r) &= \frac{A}{\rho_0 c} \frac{e^{-jkr}}{r} \left(1 + \frac{1}{jkr} \right) \end{aligned} \quad (6.9)$$

We can also directly get the Specific acoustic impedance for the (outgoing) spherical wave,

$$Z_{s,\text{spherical,outgoing}} = \rho_0 c \frac{jkr}{jkr + 1} \quad (6.10)$$

This expression for Z_s tells us, among other things, that the further away from the source of a spherical wave that we are, that is, the larger kr is, the closer is the acoustic impedance of a spherical wave to that of a plane wave (which propagates in the positive x -direction):

$$Z_{s,\text{spherical,outgoing}} = \rho_0 c \frac{jkr}{jkr + 1} \xrightarrow{kr \rightarrow \infty} \rho_0 c = Z_{s,\text{plane}}^+$$

6.3 Radiation of a spherical sound wave

In the same way as for plane waves, a spherically shaped surface with a known vibration velocity will generate an outgoing wave, and the particle velocity in the sound wave immediately close to the vibrating surface must equal the vibration velocity of the vibrating surface. We can directly look at a numerical example.

*A spherical cavity with a sound field generated inside can generate very high sound pressure amplitudes at the center, thanks to the $1/r$ -factor. In fact, the amplitude can become so high that the phenomenon of *sonoluminescence* is generated, whereby a high-amplitude sound wave generates a short (picoseconds) light pulse.

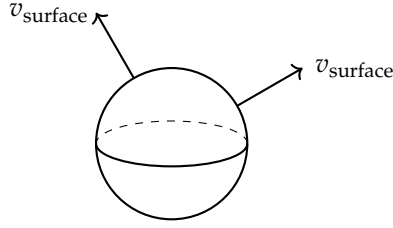


Figure 6.1: Illustration of a vibrating sphere.

Example 4: Radiation from a spherical sound source

We have a loudspeaker which is constructed as a spherical object, of radius $a = 5$ cm, the surface of which vibrates radially, with the same vibration everywhere on its surface, see Fig. 6.1. Thus, this sound source generates a spherical sound wave, that is, it radiates as a monopole.

We can make this sound source radiate at any frequency between 100 Hz and 1 kHz. The sound pressure amplitude at a distance of 1 m from the center of the spherical sound source is measured to have the value $p_{\text{rms}} = 0.1$ Pa for any frequency in the range 100 Hz - 1 kHz.

What is the *surface vibration velocity and vibration amplitude* for this source?

Calculate this for the frequencies 100 Hz, 200 Hz, 400 Hz, 800 Hz, 1kHz.

Solution:

Since it is stated that we have a spherical outgoing wave, we can assume that the solution must have this form

$$p(r) = A \frac{e^{-jkr}}{r},$$

$$v_r(r) = \frac{A}{\rho_0 c} \frac{e^{-jkr}}{r} \left(1 + \frac{1}{jkr} \right)$$

We know p_{rms} for $r = r_{\text{meas.}} = 1$ m, so:

$$p_{\text{rms}}(r_{\text{meas.}}) = \frac{|p(r_{\text{meas.}})|}{\sqrt{2}} = \frac{A/r_{\text{meas.}}}{\sqrt{2}}$$

$$\Rightarrow A = \sqrt{2} r_{\text{meas.}} \cdot p_{\text{rms}}(r_{\text{meas.}})$$

Then we can insert this value for A in the expression for the vibration

velocity for $r = a = 5$ cm:

$$\begin{aligned} v_r(a) &= \frac{A}{\rho_0 c} \frac{e^{-jka}}{a} \left(1 + \frac{1}{jka} \right) \\ &= \frac{\sqrt{2}a \cdot p_{\text{rms}}(r_{\text{meas.}})}{\rho_0 c} \frac{e^{-jka}}{a} \left(1 + \frac{1}{jka} \right) \end{aligned} \quad (6.11)$$

We are (for now) only interested in the amplitude of the vibration velocity, so we get the magnitude of v_r :

$$|v_r(a)| = \frac{\sqrt{2}r_{\text{meas.}} \cdot p_{\text{rms}}(r_{\text{meas.}})}{\rho_0 c a} \sqrt{1 + \frac{1}{(ka)^2}}$$

Finally, the vibration amplitude, $\xi_{\text{surface}} = v_r(a)/(j\omega)$ Numerically, we get (with $r_{\text{meas.}} = 1$ m, $p_{\text{rms}}(r_{\text{meas.}}) = 0.1$ Pa, $\rho_0 c = 1.2 \cdot 343$ kg m⁻² s⁻², $a = 0.05$ m, and $k = 2\pi f/c$,
 $|v_r(a)| = 75.3, 38.1, 20.0, 11.6, 10.2$ mm/s.

The vibration amplitude will then be

$$|\xi_{\text{surface}}| = \frac{|v_r(a)|}{2\pi f}$$

which numerically gives

$$|\xi_{\text{surface}}| = 120, 30, 7.9, 2.3, 1.6 \text{ } \mu\text{m}.$$

6.4 Monopoles vs dipoles

The example in the previous section studied a monopole source. Many sound sources behave as monopoles at low frequencies, even if they are not shaped as a sphere, and vibrate symmetrically. You have possibly experienced that if you dip an object of *any shape* into a calm water surface, a spherical water wave will spread outwards. A loudspeaker typically has a rectangular box with six rigid surfaces, and in one of the surfaces, a circular loudspeaker membrane is vibrating. Immediately close to the surface of the box, the particle velocity in the sound field will follow exactly the vibration velocity of the surface: zero on five surfaces, and parts of the sixth, and some vibration amplitude right at the circular membrane. But, as we move away from the surface, the wave will be closer and closer to a spherical wave! The lower the frequency is, the more spherical will the radiated wave be.

The reason for this transition towards a spherical wave is that when the vibrating surface pushes outwards, the pressure is increased locally in front of the vibrating surface, but the increased pressure wants to reach the equilibrium, the static air pressure, so the air particles *push in all directions* and thereby create a wave propagation in all directions, also around the corners.

If the vibrating surface is vibrating slowly enough (that is, with a low enough frequency), the air pressure reaction has time to spread/propagate before the vibrating surface causes a decreased pressure in front instead.

How can one "fit" a solution with the unknown constant A , as in Example 4 above to a vibrating surface of a non-spherical shape, and with a non-constant vibration velocity along its surface? It is beyond the scope of this course, but one has to use the more complicated model in Eq. (6.6) to create a more detailed soundfield - but the so-called higher-order modes, with $n > 0$, will be *evanescent*, that is, die out, as r increases, and the first monopole term is the only one that prevails.

It turns out that a sound source of monopole type gives a sound pressure which is related to the so-called *volume velocity* of the sound source, U_0 , with the unit m^3/s , through this general relationship

$$p(r) = \frac{j\omega\rho_0 U_0}{4\pi} \frac{e^{-jkr}}{r} \quad (6.12)$$

The quantity U_0 is often called *Source strength*. The volume velocity is the product of the vibration velocity of the source, multiplied with the area of the vibrating surface - regardless of the shape. The derivations in Example 4 above actually agree perfectly with Eq. (6.12). If we start from the expression in Eq. (6.11), we can find an expression for $U_0 = 4\pi a^2 v_r(a)$, then use the low-frequency assumption which implies that $ka \ll 1$, and arrive at an expression for $p(r_{\text{meas.}})$ which is compatible with Eq. (6.12) (we have to insert a factor e^{-jka}).

6.4.1 The dipole source

It was mentioned in the introduction that the simplest type of source for EM waves is a so-called *dipole source*. Such a source can be modeled/constructed/viewed as two monopole sources that are placed very close to each other, with the same source strength *but opposite polarities*. A practical example to illustrate this is a loudspeaker membrane which is not mounted in a loudspeaker box. When the membrane is moving forward, it creates a local overpressure in front of the vibrating membrane at the same time as a local *underpressure* is created at the back of the vibrating membrane. Such an unmounted loudspeaker element radiates like a dipole source. We will see now what the radiated sound pressure can be from two such monopoles.

As illustrated in Fig. 6.2, two monopoles with source strengths $+U_0$ and $-U_0$ are placed in coordinates $(x, y, z) = (0, 0, \frac{b}{2})$ and $(0, 0, -\frac{b}{2})$, respectively. If a receiver is placed in $(x_R, y_R, z_R) = (r \sin \theta, 0, r \cos \theta)$, then we can derive the

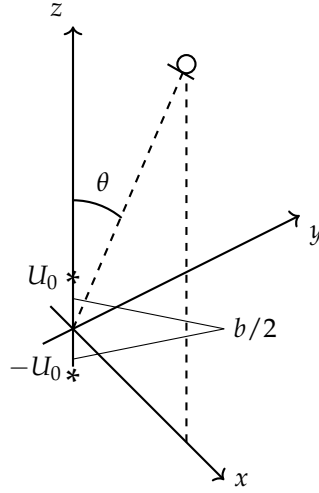


Figure 6.2: Illustration of a dipole source, implemented as two monopoles with source strengths U_0 and $-U_0$, respectively.

two distances from the two sources to the receiver:

$$r_+ = \sqrt{x_R^2 + \left(z_R - \frac{b}{2}\right)^2} = \sqrt{(r \sin \theta)^2 + \left(r \cos \theta - \frac{b}{2}\right)^2} = \sqrt{r^2 - br \cos \theta + \frac{b^2}{4}}$$

$$r_- = \sqrt{x_R^2 + \left(z_R + \frac{b}{2}\right)^2} = \sqrt{(r \sin \theta)^2 + \left(r \cos \theta + \frac{b}{2}\right)^2} = \sqrt{r^2 + br \cos \theta + \frac{b^2}{4}}$$

Now, a dipole is assumed to have the two monopoles very close to each other, so that $b \ll r$ and then we can simplify these two expressions somewhat - we find out that the last term inside the square root is much smaller than the other, and can furthermore use that $\sqrt{1+x} \approx 1+x/2$ for small x ,

$$b \ll r \Rightarrow \begin{cases} r_+ \approx r \sqrt{1 - \frac{b \cos \theta}{r}} \approx r \left(1 - \frac{b \cos \theta}{2r}\right) = r - \frac{1}{2}b \cos \theta \\ r_- \approx r \sqrt{1 + \frac{b \cos \theta}{r}} \approx r \left(1 + \frac{b \cos \theta}{2r}\right) = r + \frac{1}{2}b \cos \theta \end{cases}$$

Now we can write down an expression for the sound pressure as a sum of two monopole sound fields (such a sound source construction is sometimes called a "doublet source", as a realization of a dipole source):

$$\begin{aligned} p_{\text{doublet}}(r) &= \frac{j\omega U_0}{4\pi} \left[\frac{e^{-jk r_+}}{r_+} - \frac{e^{-jk r_-}}{r_-} \right] \\ &= \frac{j\omega U_0}{4\pi} \left[\frac{e^{-jk \left(r - \frac{b \cos \theta}{2}\right)}}{r \left(1 - \frac{b \cos \theta}{2r}\right)} - \frac{e^{-jk \left(r + \frac{b \cos \theta}{2}\right)}}{r \left(1 + \frac{b \cos \theta}{2r}\right)} \right] \end{aligned}$$

Now we can break out a spherical spreading factor e^{-jkr}/r from inside the square brackets and get

$$\begin{aligned} p_{\text{doublet}}(r) &= \frac{j\omega U_0}{4\pi} \frac{e^{-jkr}}{r} \left[\frac{e^{jk\frac{b\cos\theta}{2}}}{1 - \frac{b\cos\theta}{2r}} - \frac{e^{-jk\frac{b\cos\theta}{2}}}{1 + \frac{b\cos\theta}{2r}} \right] \\ &= \frac{j\omega U_0}{4\pi} \frac{e^{-jkr}}{r} \frac{e^{jk\frac{b\cos\theta}{2}} \left(1 + \frac{b\cos\theta}{2r}\right) - e^{-jk\frac{b\cos\theta}{2}} \left(1 - \frac{b\cos\theta}{2r}\right)}{1 - \frac{b^2\cos^2\theta}{4r^2}} \end{aligned}$$

In the denominator, the term $\frac{b^2\cos^2\theta}{4r^2}$ will be much smaller than 1, so we can ignore that second term:

$$\begin{aligned} p_{\text{doublet}}(r) &\approx \frac{j\omega U_0}{4\pi} \frac{e^{-jkr}}{r} \left[e^{jk\frac{b\cos\theta}{2}} \left(1 + \frac{b\cos\theta}{2r}\right) - e^{-jk\frac{b\cos\theta}{2}} \left(1 - \frac{b\cos\theta}{2r}\right) \right] \\ &= \frac{j\omega U_0}{4\pi} \frac{e^{-jkr}}{r} \left[e^{jk\frac{b\cos\theta}{2}} - e^{-jk\frac{b\cos\theta}{2}} + \frac{b\cos\theta}{2r} \left[e^{jk\frac{b\cos\theta}{2}} + e^{-jk\frac{b\cos\theta}{2}} \right] \right] \\ &= \frac{j\omega U_0}{4\pi} \frac{e^{-jkr}}{r} \left[2j \sin\left(\frac{kb\cos\theta}{2}\right) + \frac{b\cos\theta}{2r} 2 \cos\left(\frac{kb\cos\theta}{2}\right) \right] \end{aligned}$$

Now we use a series expansion, since we can assume that $kb \ll 1$, and $\sin x \approx x$ while $\cos x \approx 1$ for small x , so

$$\begin{aligned} p_{\text{doublet}}(r) &\approx \frac{j\omega U_0}{4\pi} \frac{e^{-jkr}}{r} \left[2j \frac{kb\cos\theta}{2} + \frac{b\cos\theta}{2r} 2 \right] \\ &= \frac{j\omega U_0}{4\pi} \frac{e^{-jkr}}{r} jkb\cos\theta \left(1 + \frac{1}{jkr}\right) = p_{\text{dipole}} \\ &\Rightarrow p_{\text{dipole}} = p_{\text{monopole}} \cdot jkb\cos\theta \left(1 + \frac{1}{jkr}\right) \end{aligned}$$

What does this last expression tell us? It tells us:

- The dipole source is directional - it has a $\cos\theta$ -factor, which implies that the field from a dipole source is zero in some directions, and with opposite polarity in the forward and rear directions.
- The dipole source has a near-field (when $kr \ll 1$) and a far-field (when $kr \gg 1$), with different responses.
- In the far-field, the dipole source has a factor $k \sim \omega$ compared to the monopole source - so the low-frequency radiation from a dipole is much weaker than from a monopole with the same source strength U_0 . This is the reason that loudspeaker elements are placed in boxes: to turn the dipole into a monopole, which gives much more bass output.

Often, the sound field from a dipole uses the so-called "Dipole moment", D_0 , as source strength descriptor: $D_0 = bU_0$ and then

$$p_{\text{dipole}} = -\frac{\omega^2 D_0}{4\pi c} \frac{e^{-jkr}}{r} \cos \theta \left(1 + \frac{1}{jkr} \right)$$

As mentioned earlier, dipole sources are the simplest types of sources of EM waves, and thus, dipole sources will be encountered in the part on EM waves.